THARANEE.DEV

PRODUCT ENGINEERING DOCUMENTATION

PRODUCT TECHNICAL DOCUMENT

Cartizion Bot V1.1

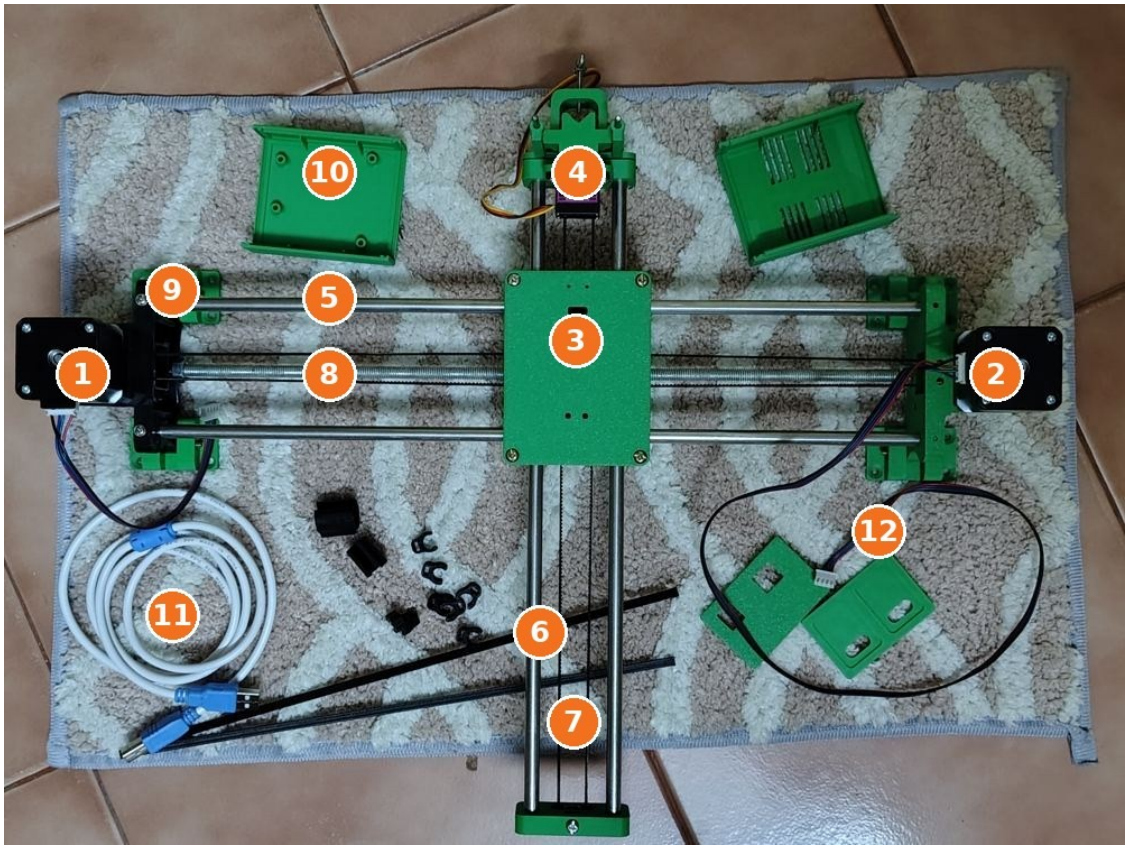
Pen plotter and writing machine

Products by Tharanee.dev

225 × 320 mm	EBB v3.0.3	12 V 2 A
Drawing area (A4)	Controller firmware	Power supply

Field	Entry
Product	Cartizion Bot V1.1
Brand	Products by Tharanee.dev
Document number	CB-TD-001
Revision	A (draft)
Date	29 October 2026
Classification	Product technical reference

Main View - Cartizion Bot V1.1



MAIN VIEW *Cartizion Bot V1.1 with numbered parts (products by Tharanee.dev)*

1	Motor A (left)	2	Motor B (right)
3	Carriage plate	4	Pen head with MG90S servo
5	X guide rods	6	Y guide rods
7	Timing belts (Y axis)	8	Threaded rod (function: verify)
9	Rod end block / motor bracket	10	Printed cover (spare)
11	USB cable	12	Motor lead and spare parts

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1. Introduction and Product Overview

1.1 Purpose and Scope

This document is the product technical reference for the Cartizion Bot V1.1, a pen plotter (writing and drawing machine) by Tharanee.dev. It specifies the machine, its mechanical and electrical components, the EBB control board, stepper motors and drivers, the assembly procedure, software installation, pre-checks, commissioning, operation, maintenance, and project summary, with appendices for BOMs, drawings, schematics, datasheets, calculations and test reports.

1.2 Product Description

The Cartizion Bot V1.1 writes or draws on almost any flat surface using pens, markers, pencils and other writing tools. It is controlled from Inkscape with the AxiDraw extension over a single USB cable and uses an EBB-compatible controller running EBB firmware v3.0.3. The machine is supplied fully assembled and ready to use out of the box.

The writing head extends beyond the body of the machine, so it can draw on objects larger than the machine itself and the machine can be placed over a book. The structure is printed in ABS plastic, with injection-moulded parts where strength matters.

1.3 Key Features

- EBB-compatible controller running EBB firmware v3.0.3, with USB firmware updates
- Works with Inkscape 1.x and the AxiDraw extension (free software)
- Drawing area 225 × 320 mm, A4 paper capacity
- Holds pens up to 15 mm in diameter
- Servo pen lift, 5 V
- Removable StepStick driver sockets: A4988, TMC2208 or TMC2209 (near-silent with TMC2208 / TMC2209)
- 12 V DC supply (12 to 24 V board input range)
- 3D-printed ABS parts, printed in a 70 °C heated chamber, plus injection-moulded parts
- Plug and play: connect USB and the power adapter, install the software and start drawing

1.4 What Is in the Box

Package	Contents
Cartizion Bot V1.1 machine	1 × Cartizion Bot V1.1 writing machine (fully assembled); 1 × 12 V 2 A power adapter; 1 × USB cable (Type-A to Type-B)
Cartizion Bot EBB Control Board (stand-alone)	1 × EBB control board. Stepper drivers, motors, servo, USB cable and power adapter are not included.

1.5 Warranty, Delivery and Support

Item	Terms [Confirm before issue]
Warranty (machine)	3 months. Damaged or faulty parts are replaced with guidance for installation; if the machine cannot be repaired, it is replaced.

Item	Terms [Confirm before issue]
After warranty	Full support continues. All spare parts are available for purchase.
Delivery	5 to 8 working days anywhere in India
Price	Includes GST
Board only	Tested before shipping; replaced if damaged in shipping; no other warranty
Support contact	[Tharanee.dev support contact: phone / WhatsApp / e-mail]

1.6 Conventions

NOTE Tags: [Spec] = product specification. [Photo] = identified from the build photographs. [Verify] = typical value for this class of component; confirm against the part marking or datasheet. Product name: Cartizion Bot V1.1, products by Tharanee.dev.

2. Technical Specifications

2.1 Machine Specifications

Parameter	Specification
Drawing area	225 × 320 mm (maximum paper size A4)
Writing speed	One A4 page of cursive text takes about 6 to 8 minutes; non-cursive fonts take 8 to 20 minutes. Speed depends on the font type.
Maximum travel speed	330 mm/s [from earlier specification table]
Pen size	Holds pens up to 15 mm diameter
Machine size	56 × 40 × 12 cm
Power	12 V 2 A adapter (included)
Connection	USB Type-B (Type-A to Type-B cable included)
Controller	Cartizion Bot EBB Control Board, EBB firmware v3.0.3
Software	Inkscape 1.4.4 + AxiDraw extension (free)
Computers	Windows 10 or newer, macOS 10.15 Catalina or newer
Build	3D-printed ABS parts and injection-moulded parts
Assembly	Fully assembled, ready to use out of the box

Drawing area	225 × 320 mm (fits A4 paper)
Maximum speed	330 mm per second
Pen size	Holds pens up to 15 mm diameter
Machine size	56 × 40 × 12 cm
Power	12 V 2 A adapter (included)
Connection	USB Type-B
Controller	EasyDraw EBB Control Board, EBB firmware v3.0.3
Software	Inkscape + AxiDraw extension (free). Tested with Bantam Tools NextDraw™ Merge (sold separately)
Computers	Windows 10 or newer, macOS 10.15 Catalina or newer
Assembly	Fully assembled, ready to use

FIGURE 1 *Source specification table*

2.2 Computer and Software Requirements

Item	Requirement
Operating system	Windows 10 / 11, or macOS 10.15 Catalina or newer

Item	Requirement
Software	Inkscape 1.x with the AxiDraw extension (free)
Port	One free USB-A port on the computer
Internet	Needed once to download the software

2.3 System Block Diagram

The diagram shows how data and power flow from the computer to the pen head. Numbers 1 to 6 follow the signal path.

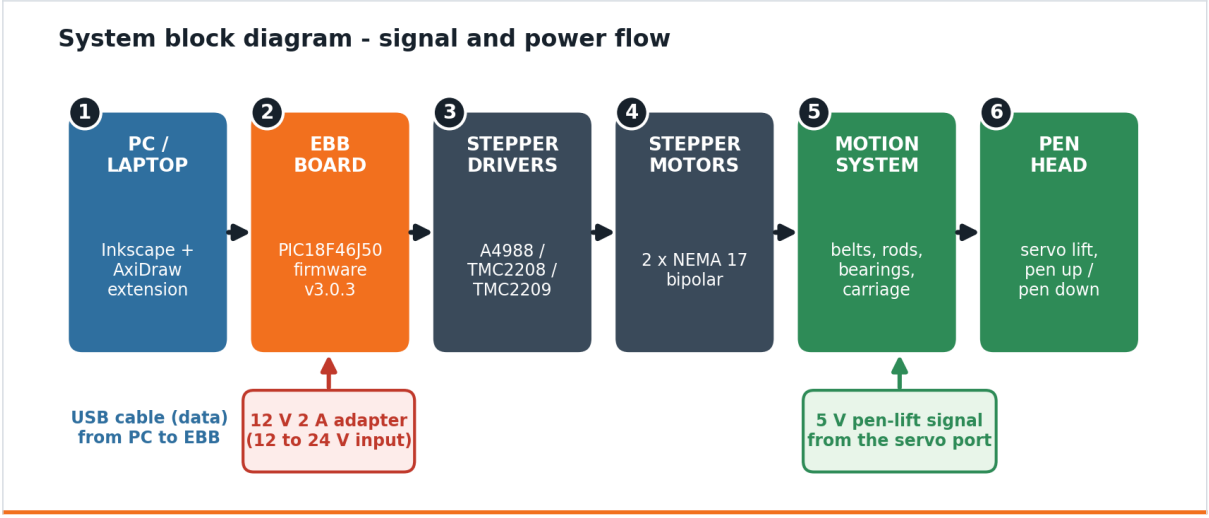


FIGURE 2 *System block diagram: signal and power flow*

3. Mechanical Specification

3.1 Machine Architecture

The machine uses a CoreXY-type belt drive: two stepper motors at the two ends of the long axis move the carriage in X and Y together. The writing head is carried on the front of the carriage and projects beyond the machine body. The controller supports both CoreXY and Cartesian (X/Y) machines.

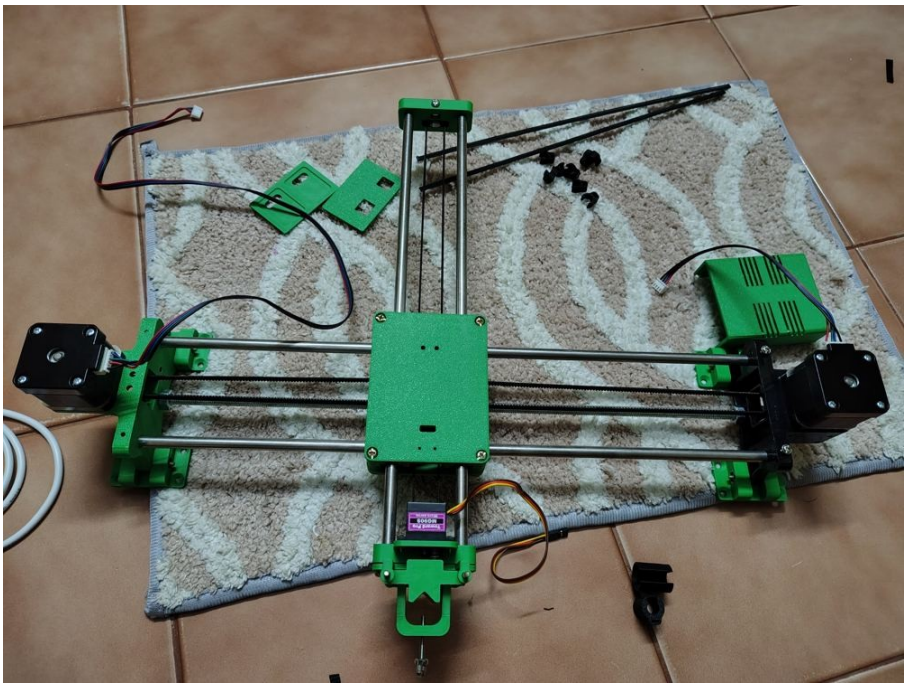


FIGURE 3 *Machine frame with both motors, rods and carriage*

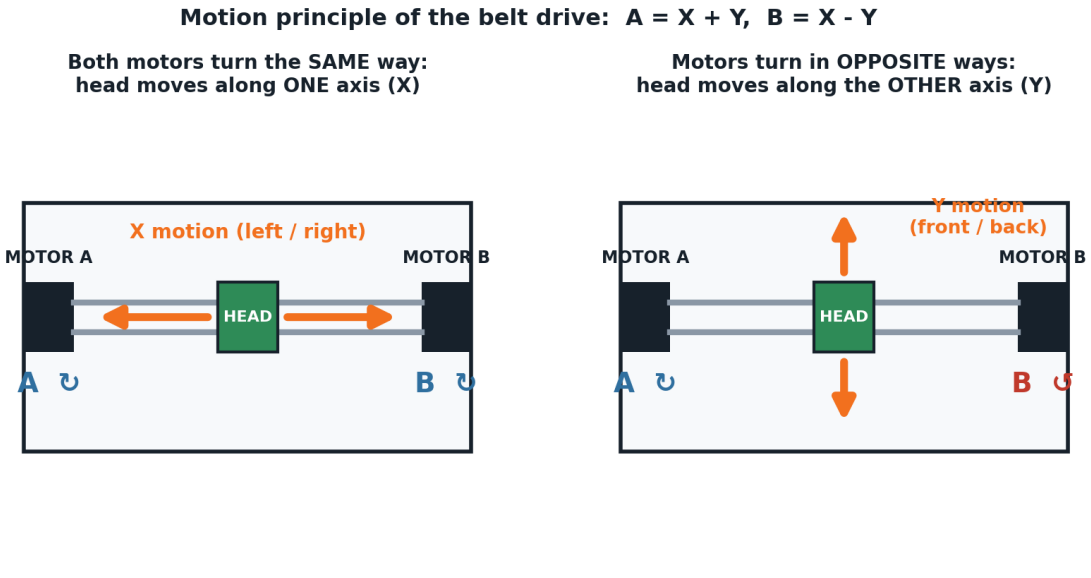


FIGURE 4 *Motion principle of the belt drive*

3.2 Structural Parts and Material

All 3D-printed parts are made in ABS plastic in a 70 °C heated chamber, which gives strong layer bonding. ABS was chosen over PLA for these reasons:

Property	PLA	ABS (used)
Strength over time	Brittle; loses strength after some months	Tough and impact resistant
Heat resistance	Low glass transition temperature; deforms in sun or a parked car	Holds shape at higher temperature
Printing	Easy	Needs an actively heated, enclosed printer



FIGURE 5 *Upper and lower carriage plates before assembly*

Property	Requirement
Printed material	ABS, printed in a 70 °C heated chamber [Spec]
Other parts	Injection-moulded parts [Spec]
Rod bores	Light sliding fit for the rod diameter
Fastening	Machine screws and nuts through printed holes

3.3 Guide Rods and Linear Bearings



FIGURE 6 *Guide rods with linear bearings pre-fitted*

Property	Specification
Rod type	Smooth ground steel [Photo]
Rod diameter	8 mm typical [Verify]
Arrangement	Two parallel rods per axis, perpendicular between axes
Bearings	Four on the carriage per rod pair [Photo]; eight laid out in the kit
Threaded rod	Long threaded rod beside the X rods [Photo]; function to be confirmed [Verify]
Straightness	Roll each rod on a flat surface; no bends or burrs

3.4 Belt and Pulley Drive

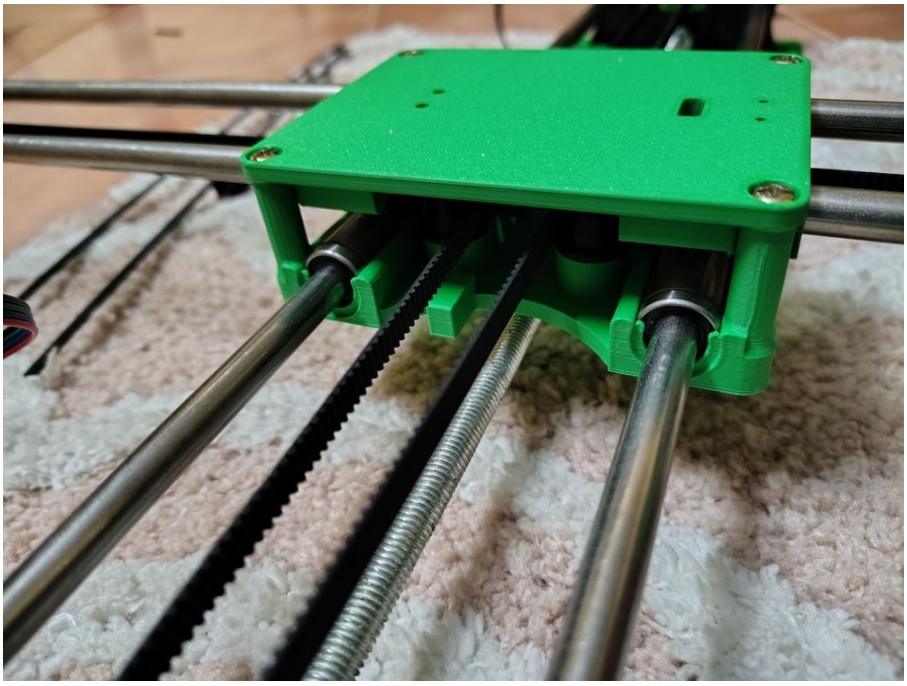


FIGURE 7 *Belt run and rod end block (close-up)*

Property	Specification
Belt	Black toothed timing belt [Photo]
Typical pitch	GT2, 2 mm [Verify]
Pulleys	20-tooth motor pulleys typical [Verify]; idlers at the far ends [Photo]
Travel per motor revolution	40 mm (20 teeth × 2 mm) [Verify]
Tension	Taut; 1 to 2 mm deflection under light finger pressure

3.5 Pen-Lift Mechanism

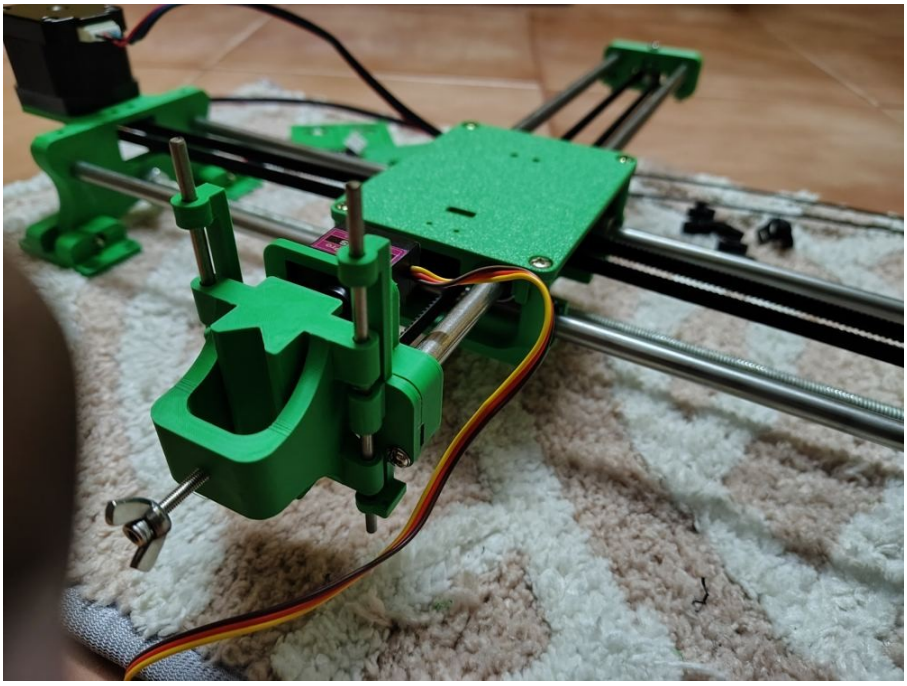


FIGURE 8 *Pen head with servo mounted on the carriage*

Property	Specification
Actuator	MG90S-type metal-gear micro servo, 5 V [Photo: label MG90S]
Pen capacity	Up to 15 mm diameter [Spec]
Retention	Printed holder with thumb screw
Travel	Pen-up and pen-down heights set in the software

3.6 Overall Dimensions

Parameter	Value
Machine size	56 × 40 × 12 cm
Drawing area	225 × 320 mm
Paper	A4 (210 × 297 mm) fits inside the drawing area
Placement	Can be placed over a book; the head extends beyond the machine

4. Electrical Specification

4.1 Electrical Summary

Parameter	Value
Supply voltage	12 V DC (board input range 12 to 24 V)
Supply current	2 A (12 V adapter)
Supply power	24 W maximum at 12 V
Power connector	DC barrel jack, 5.5 × 2.1 mm
Communication	USB Type-B to host PC
Control logic	EBB board, firmware v3.0.3
Motor drive	2 × StepStick drivers (A4988, TMC2208 or TMC2209)
Pen lift	5 V servo output

4.2 Power Supply

Item	Specification
Machine adapter	12 V DC, 2 A, regulated, barrel plug (included)
Alternative	24 V 3 A adapter is supported by the board for 24 V builds
Polarity	Centre positive (confirm on the adapter label)
Rule	Use the same Vref setting at 12 V and 24 V

4.3 Cables and Connectors

Cable	Connector	Notes
USB	Type-A to Type-B	Data-capable, under 2 m
Power	5.5 × 2.1 mm barrel	12 V 2 A
Stepper motors	JST-XH 4-pin	One per motor
Servo	3-pin S / + / – header	5 V pen lift

4.4 Electrical Block Schematic

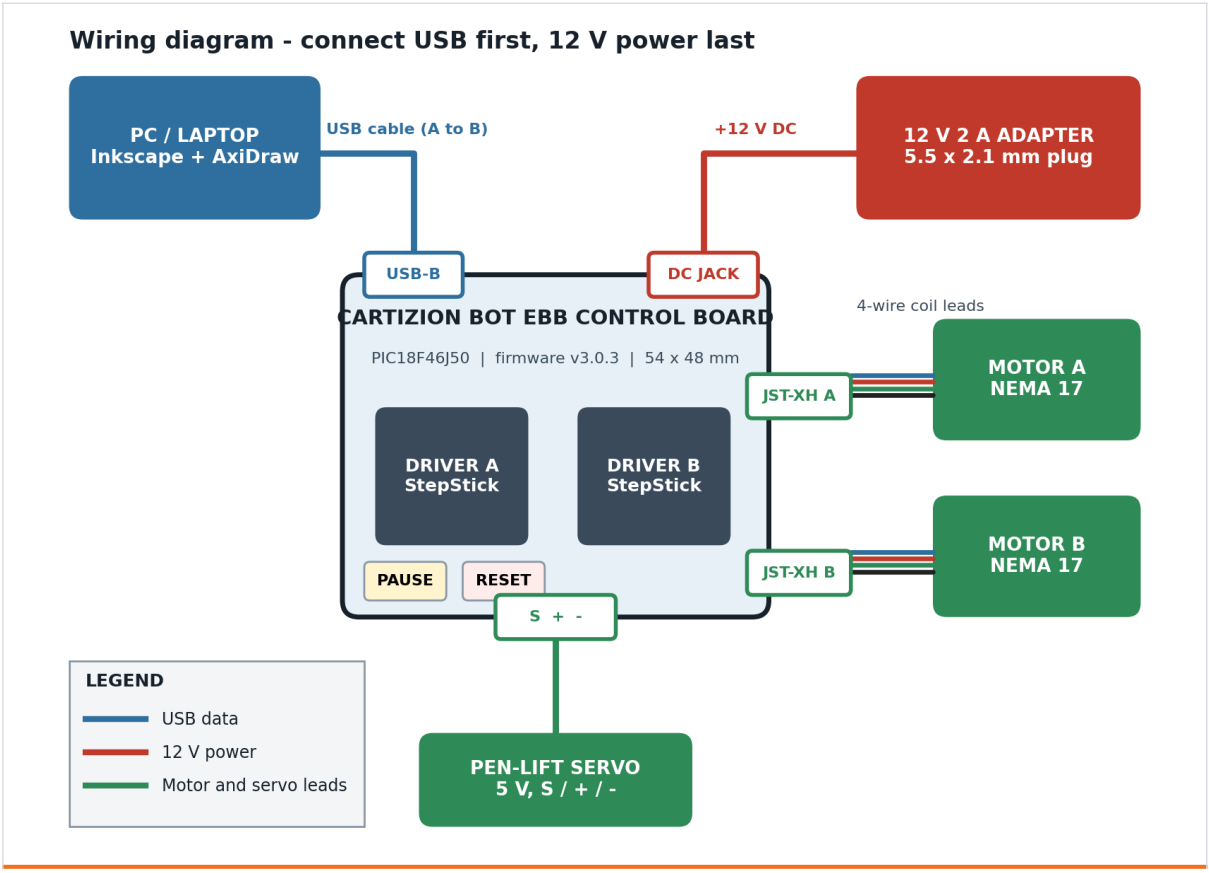


FIGURE 9 *Wiring diagram: PC, adapter, EBB board, motors and servo*

5. EBB Control Board

5.1 Key Features

- EiBotBoard (EBB) compatible, running EBB firmware v3.0.3
- Compatible with AxiDraw software: Inkscape 1.x + AxiDraw extension
- 2 removable StepStick driver sockets: A4988, TMC2208 or TMC2209
- Near-silent plotting with TMC2208 / TMC2209 drivers
- 12 to 24 V DC input
- 5 V servo output for pen lift
- Pause and reset buttons
- Firmware updates over USB, no programmer needed
- Compact 54 × 48 mm board

5.2 Technical Specification

Parameter	Value
Microcontroller	PIC18F46J50
Firmware	EBB (EiBotBoard) firmware v3.0.3, updatable over USB
Stepper drivers	2 × StepStick sockets for A4988, TMC2208, TMC2209 (not included)
Microstepping	1/16 (High resolution) or 1/8 (Low resolution), set in software
Stepper motors	2 × NEMA 17 bipolar
Motor connectors	JST-XH 4-pin
Pen-lift output	1 × servo port, 5 V
Input voltage	12 to 24 V DC (use 12 V 2 A or 24 V 3 A adapter)
Power connector	DC barrel jack, 5.5 × 2.1 mm
USB	USB Type-B
Buttons	Pause, Reset
Machine types	CoreXY, Cartesian (X/Y)
Software	Inkscape 1.x + AxiDraw extension
Computers	Windows 10 / 11, macOS (latest)
Board size	54 × 48 mm
Weight	25 g (80 g packed)
Made in	India

5.3 Compatibility

- Software: Inkscape 1.x with the AxiDraw extension.
- Computers: Windows 10 / 11 and macOS.

- Machines: Cartizion Bot V1.1, and DIY CoreXY or Cartesian pen plotters with 2 × NEMA 17 motors and a 5 V servo pen lift.

5.4 Wiring Summary

Connect the USB port to the computer, the power adapter to the barrel jack, the two stepper motors to the JST-XH ports, and the pen-lift servo to the 3-pin S / + / − header.

6. Stepper Motors and Drivers

6.1 Stepper Motor Specification



FIGURE 10 *Stepper motor fitted to its printed bracket*

The machine uses two NEMA 17 bipolar stepper motors [Spec]. Typical values for a 1.0 to 1.5 A class motor are given below; read the label on the fitted motor and record the actual values in Appendix 15.7.

Parameter	Typical value [Verify]
Type	2-phase hybrid bipolar stepper, 4 lead wires
Frame size	NEMA 17, 42.3 × 42.3 mm face
Body length	34 to 48 mm
Step angle	1.8° (200 steps per revolution), accuracy ±5% of a step
Rated current	1.0 to 1.5 A per phase
Coil resistance	1.5 to 3.5 Ω per phase
Coil inductance	2 to 6 mH per phase
Holding torque	0.35 to 0.55 N·m (35 to 55 N·cm)
Shaft	5 mm diameter, D-cut, about 22 mm long
Mounting	4 × M3 holes on a 31 mm square; 22 mm pilot boss
Insulation / temperature	Class B (130 °C); casing rise up to 80 K
Connector	JST-XH 4-pin plug to the board
Operating current setting	70 to 80% of rated current (Section 6.4)

6.2 Stepper Driver Compatibility

The board has removable driver sockets with the standard StepStick footprint. All drivers below work at both 12 V and 24 V supply.

Driver	Plug and play	Resolution setting	Notes
A4988	Yes	High or Low	Works in both modes
TMC2208	Yes	High (default)	Near-silent, smoothest motion
TMC2209	Yes	High (default)	Near-silent; some brands run the louder spreadCycle mode but still draw correctly

NOTE TMC2208 / TMC2209 users: keep Resolution on High. On Low, drawings come out half size.

6.3 Resolution Setting

High resolution (1/16 microstepping) is the default, so most users do not need to change anything. If the setting was changed earlier, set it back:

1. In Inkscape, go to Extensions > AxiDraw Control.
2. Open the Options tab.
3. Set Resolution to High.
4. Inkscape remembers the setting for future plots.

Command-line and Python users: in axidraw_conf.py, set resolution = 1.

6.4 Driver Current Setting (Vref)

Vref is a small voltage on the driver potentiometer that sets the motor current. Use the same Vref at 12 V and 24 V.

Step 1: find the driver formula

Check the tiny sense resistor marking printed near the driver chip.

Driver	Sense resistor marking	Vref formula
A4988	R100	Vref = motor current × 0.8
A4988	R068	Vref = motor current × 0.54
TMC2208 / TMC2209	R110	Vref = motor current × 1.0

Step 2: quick reference (Vref in volts)

Motor current	A4988 (R100)	A4988 (R068)	TMC2208 / 2209
0.8 A	0.64 V	0.44 V	0.80 V
1.0 A	0.80 V	0.54 V	1.00 V

Motor current	A4988 (R100)	A4988 (R068)	TMC2208 / 2209
1.2 A	0.96 V	0.65 V	1.20 V
1.5 A	1.20 V	0.82 V	1.50 V

Use the motor rated current or a little less. A pen plotter needs very little force, so 70 to 80% of the rated current is plenty and keeps the motors and drivers cooler.

Step 3: measure and adjust

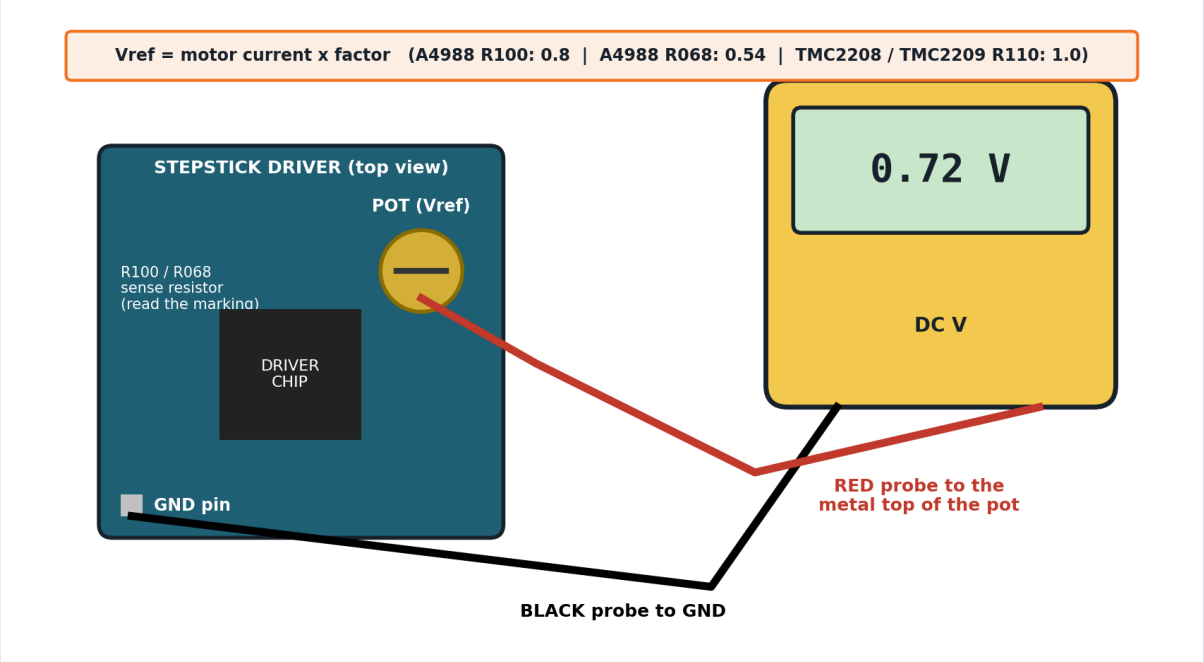


FIGURE 11 *Measuring Vref: black probe on GND, red probe on the potentiometer*

1	Power the board at 12 V or 24 V.
2	Set the multimeter to DC volts. Put the black probe on GND and the red probe on the metal top of the potentiometer.
3	Turn the potentiometer slowly with a plastic or ceramic screwdriver and re-measure until the target value is reached.

6.5 Driver Safety Rules

WARNING Always switch off power before inserting or removing a driver. Never unplug a motor while the board is powered; it can destroy the driver.

- Match the EN pin on the driver to the EN marking on the board. Some TMC modules face the opposite way from A4988 modules.
- Fit the heatsinks. Above about 1 A, add a small fan.
- If a motor runs backwards after switching drivers (the drawing comes out mirrored), reverse that motor plug or swap the two wires of one coil.

7. Pen-Lift Servo Specification

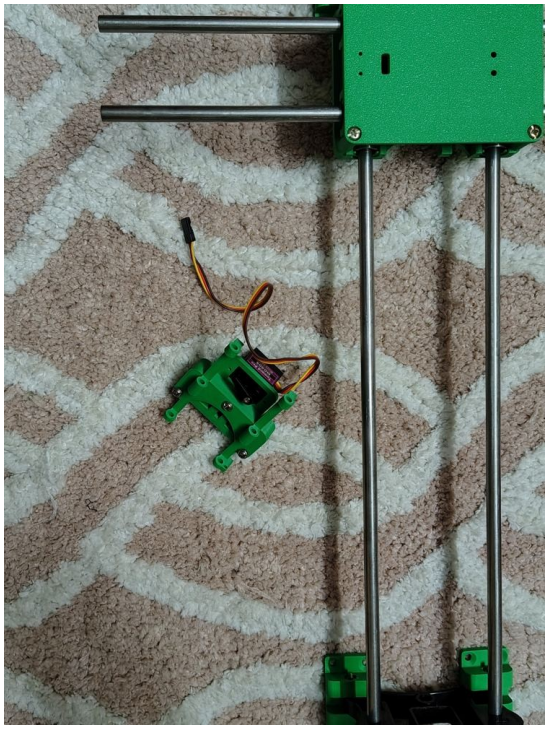


FIGURE 12 *Servo fitted in the pen head housing*

Parameter	Value
Type	MG90S-type micro servo (label visible in photo), three-wire
Supply	5 V from the board servo port [Spec]
Header	3-pin S / + / − [Spec]
Wiring	Brown or black = −, red = +, orange or yellow = S (signal)
Stall torque	About 1.8 kg·cm at 4.8 V [Verify datasheet]
Pulse range	About 1000 to 2000 µs at 50 Hz [Verify]
Pen heights	Set in AxiDraw Control, Setup tab

8. Assembly Procedure

The machine is supplied fully assembled. This procedure follows the photographed from-parts build (30 Sep 2026) and is used for assembly, rebuild after repair, and spare-part replacement. Photo time stamps are given for traceability.

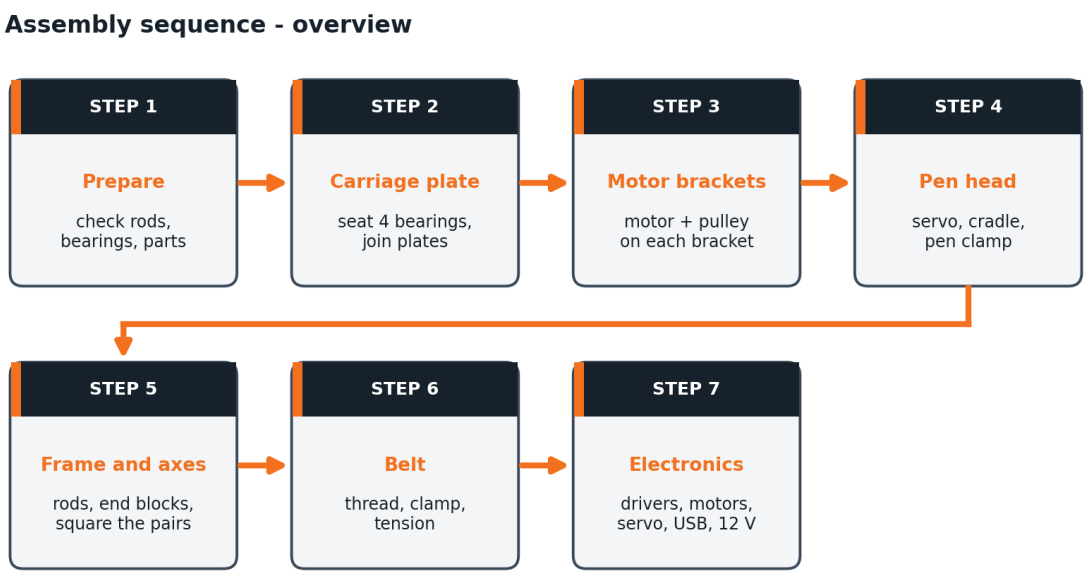


FIGURE 13 *Assembly sequence: step-by-step overview*

8.1 Preparation



FIGURE 14 *Parts kit: motors, belt, servo, USB cable, adapter and rods (22:45)*

1	Lay out all parts on a clean, flat surface and check them against the mechanical and electrical BOM (Appendix 15.1 and 15.2).
2	Check the guide rods for straightness and remove burrs from printed parts.
3	Test-fit one bearing on one rod; it should slide freely with no play.
4	Group the motors, belt, servo, cable and adapter separately.



FIGURE 15 *ABS printed parts, fasteners and black clips laid out (22:48)*

8.2 Carriage Plate Assembly

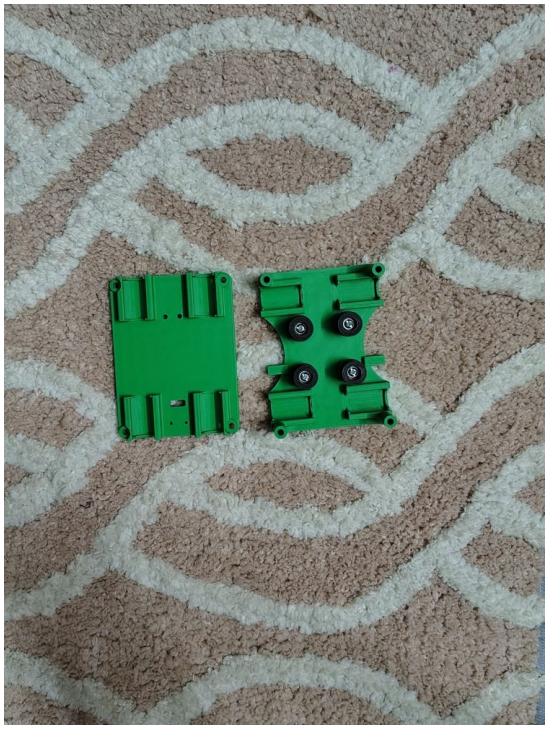


FIGURE 16 *Four bearings seated in the lower plate (22:54)*

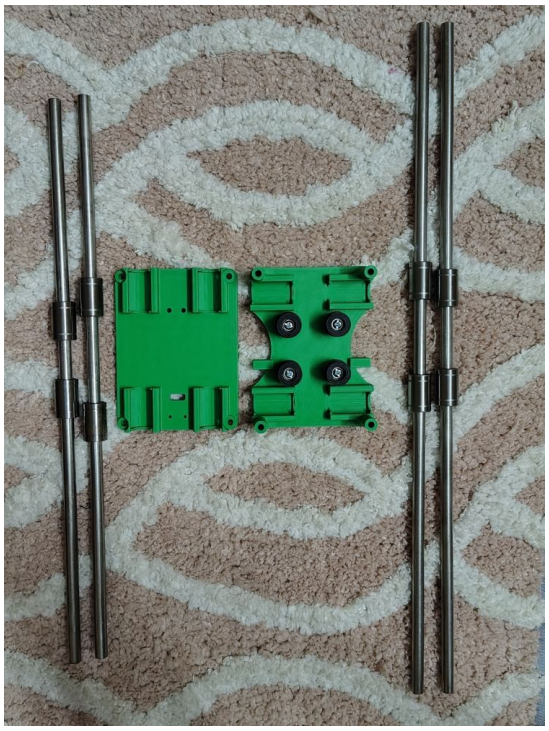


FIGURE 17 *Rods passed through the carriage bearings (22:56)*

1	Place the upper and lower carriage plates side by side.
2	Fit the black bearing clips and bearings into the lower plate pockets.

3	Seat all four bearings per axis with their axes parallel.
4	Slide the rods through the bearings, two per axis in each direction, to check alignment.
5	Join the upper plate to the lower plate with screws, tightening evenly in a cross pattern.
6	Confirm the plate slides smoothly along both rod pairs with no binding.

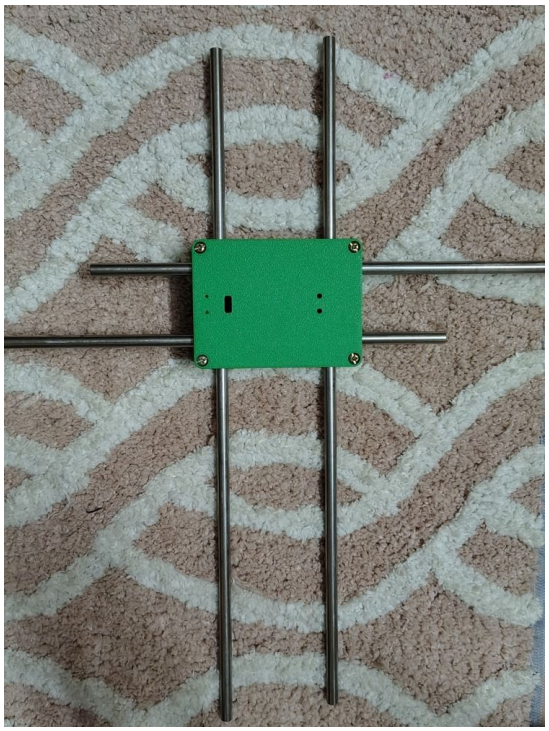


FIGURE 18 *Assembled carriage plate with four rods (23:02)*

8.3 Motor Bracket Assembly

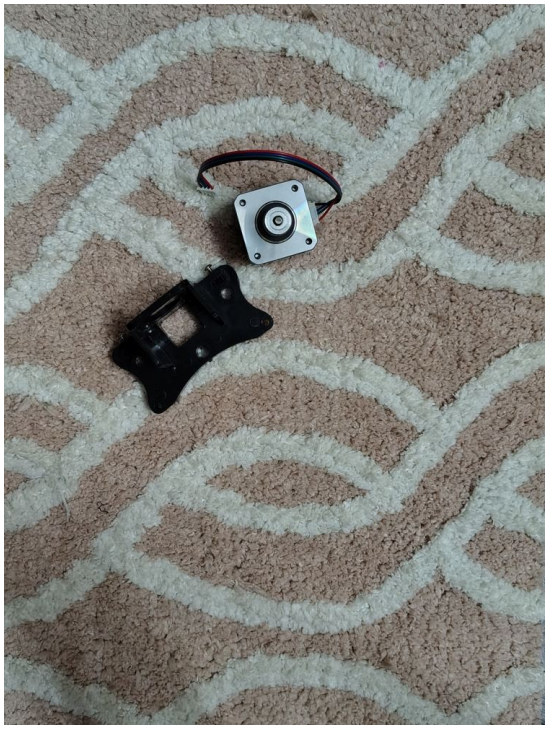


FIGURE 19 *Motor bracket parts (23:10)*

1	Fit the first motor to its printed bracket with the four face screws.
2	Press the pulley onto the motor shaft and tighten the grub screw on the flat.
3	Repeat for the second motor and bracket.
4	Fit the printed end blocks that hold the rod ends.

8.4 Pen Head Assembly

1	Fit the servo into the printed pen head housing.
2	Attach the servo horn and pen cradle; leave the horn screw loose until calibration.
3	Install the clamp and thumb screw for pen retention.
4	Route the servo lead so it cannot snag in motion.

8.5 Frame and Axis Assembly

1	Mount the motor brackets at the two ends of the long axis.
2	Slide the rods through the end blocks, then through the carriage bearings.
3	Fix the end blocks to the rods, keeping the two rod pairs square.
4	Mount the pen head on the front of the carriage.
5	Check that both axes move through full travel by hand.

8.6 Belt Installation and Tensioning

1	Thread the belt around the motor and idler pulleys.
2	Clamp the belt ends into the carriage belt holders.
3	Tension the belt until taut but not tight, with both runs level.
4	Slide the carriage by hand and check the belt tracks centrally on the pulleys.

8.7 Electronics Installation

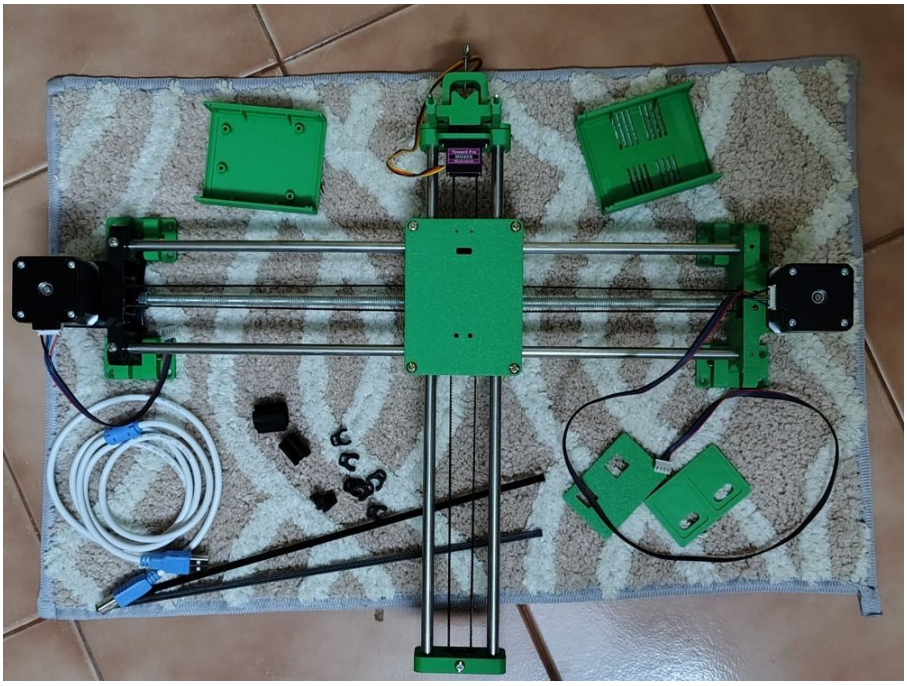


FIGURE 20 *Final layout with covers, cables and USB lead (23:49)*

1	Fit the printed covers over the electronics area.
2	Fit the stepper drivers into the board sockets with power off, matching the EN marking.
3	Connect the stepper plugs (JST-XH) and the servo plug (S / + / −) to the board.
4	Connect the USB cable between the board and the PC.
5	Connect the 12 V adapter last.
6	Secure and tidy the cables clear of the moving axes.

WARNING Do not connect or disconnect a motor while the board is powered; this can damage the stepper drivers.

9. Wiring and Connections

From	To	Notes
12 V adapter	Board DC barrel jack	Connect last
USB cable	Board USB Type-B to PC USB-A	Use a data-capable cable
Stepper motor A	Board motor port A (JST-XH)	Reverse the plug or swap one coil pair if the direction is wrong
Stepper motor B	Board motor port B (JST-XH)	As above
Servo	Board 3-pin S / + / − header	Signal, power, ground in marked order
Drivers	StepStick sockets A and B	Power off; EN marking matches

9.1 Wiring Rules

- Keep motor leads away from the USB cable.
- Do not run the machine without the drivers heatsinked.
- Check the power adapter label: 12 V DC, centre positive, 2 A minimum.
- Connect USB first and power last; disconnect power first.

10. Software Installation and Setup

10.1 Software Overview

The machine is controlled from Inkscape, a free vector graphics editor, using the free AxiDraw extension. Both are public software and are downloaded from the internet with a browser such as Google Chrome. No programming is required, and the controller ships ready: no code needs to be uploaded.

Component	Source (open in Chrome)
Inkscape	inkscape.org download page: choose the installer for Windows (exe or msi) or macOS (dmg)
AxiDraw software for Inkscape	axidraw.com/sw (redirects to the Evil Mad Scientist AxiDraw Software Installation wiki page)
USB driver (Windows)	Included in the AxiDraw Windows installer, which is needed for older Windows versions

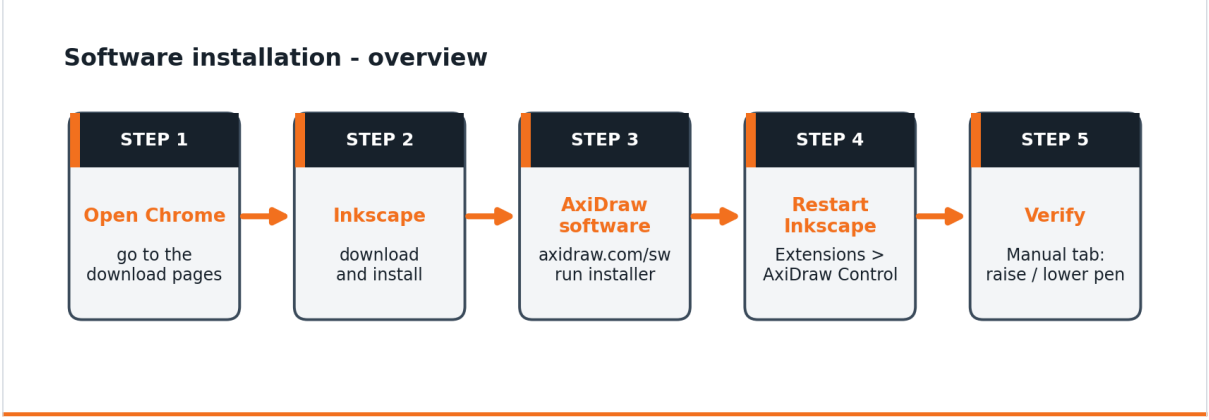


FIGURE 21 *Software installation: step-by-step overview*

10.2 Install Inkscape

1. Open Google Chrome and go to the Inkscape download page.
2. Select your operating system and download the installer. Windows: exe or msi. macOS: dmg.
3. Run the installer and follow the on-screen steps. Windows needs administrator rights.
4. Open Inkscape once to confirm it starts, then close it.

NOTE The product specification names Inkscape 1.4.4. The AxiDraw software page lists the Inkscape versions each release supports; at the time of writing it requires Inkscape 1.0 or newer and recommends 1.2 or newer. Check the page for the newest release before installing [Verify].

10.3 Install the AxiDraw Software

1. In Chrome open axidraw.com/sw and read the section for your operating system.

2	Windows: download the AxiDraw software installer (a ZIP archive), unzip it, and run "AxiDraw Install.exe".
3	macOS: download and run the installer matching your macOS version and Inkscape version listed on the page.
4	Linux users follow the manual installation instructions on the same page.
5	Restart Inkscape. The AxiDraw extension appears under Extensions > AxiDraw Control.

10.4 USB Driver (Windows)

Windows 8 and older need a USB driver for the controller. The AxiDraw Windows installer includes it, so run the installer once on those systems. Windows 10 and 11 normally recognise the board without a separate driver.

10.5 Verify the Installation

1	Connect the board to the PC with the USB cable, then power the machine with the 12 V adapter.
2	Open Inkscape and choose Extensions > AxiDraw Control.
3	Open the Manual tab and press "Raise pen" and "Lower pen"; the pen head should move.
4	If Inkscape reports that it cannot find the machine, check the USB cable, power and Section 13.8.

10.6 Command-Line and Python Option

The same software can also be run from the command line or Python. Install Python 3, then install the AxiDraw Python module as described on the AxiDraw software page (python -m pip install with the module archive link given there) [Verify current link]. This option is for automation; normal plotting uses Inkscape.

10.7 Legacy Software Note

The first-generation legacy AxiDraw (sold before February 2016, white Delrin chassis) was controlled with the WaterColorBot (WCB) extensions developed by Evil Mad Scientist, which have a "pen only" mode. These extensions run in Inkscape and are available from the Evil Mad Scientist GitHub repository, with a Windows USB driver installer (UBWDriverInstaller). Because the controller appears as a serial port, commands can also be sent from any terminal emulator using the EiBotBoard command set. This path is only for reference with older Inkscape versions; the Cartizion Bot V1.1 uses the current AxiDraw software above.

11. Stepper Motor Pre-Check

Complete this section before the first powered movement. It protects the drivers and motors and records a baseline for later maintenance.

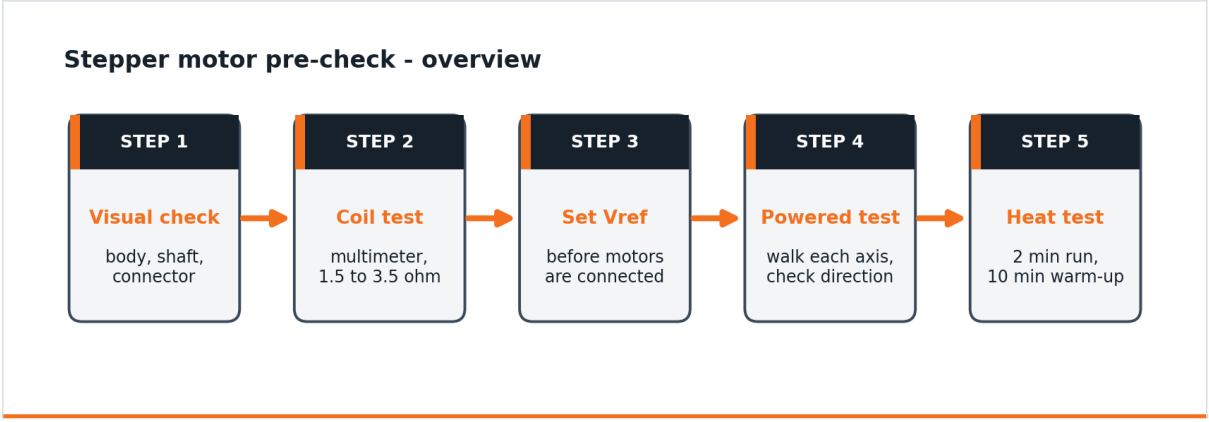


FIGURE 22 *Stepper motor pre-check: step-by-step overview*

11.1 Visual and Mechanical Check

1	Inspect each motor body, shaft and connector for damage or bent pins.
2	Rotate each shaft by hand with the motor disconnected; it should turn with light, even cogging and no grinding.
3	Check the pulley is fully seated and the grub screw sits on the shaft flat.
4	Check the motor is square to its bracket and all four screws are tight.
5	Move the carriage by hand over its full travel with the belt fitted; there must be no tight spots.

11.2 Coil Identification and Resistance Test

1	Set a multimeter to the lowest resistance range.
2	Measure between pairs of wires. A pair showing a low reading (about 1.5 to 3.5 Ω) is one coil.
3	Wires from different coils read open circuit.
4	Record the two coil resistances; they should match within about 10%.
5	Quick spin test: short the two wires of one coil together and turn the shaft by hand. Noticeably increased resistance confirms that pair is a coil.

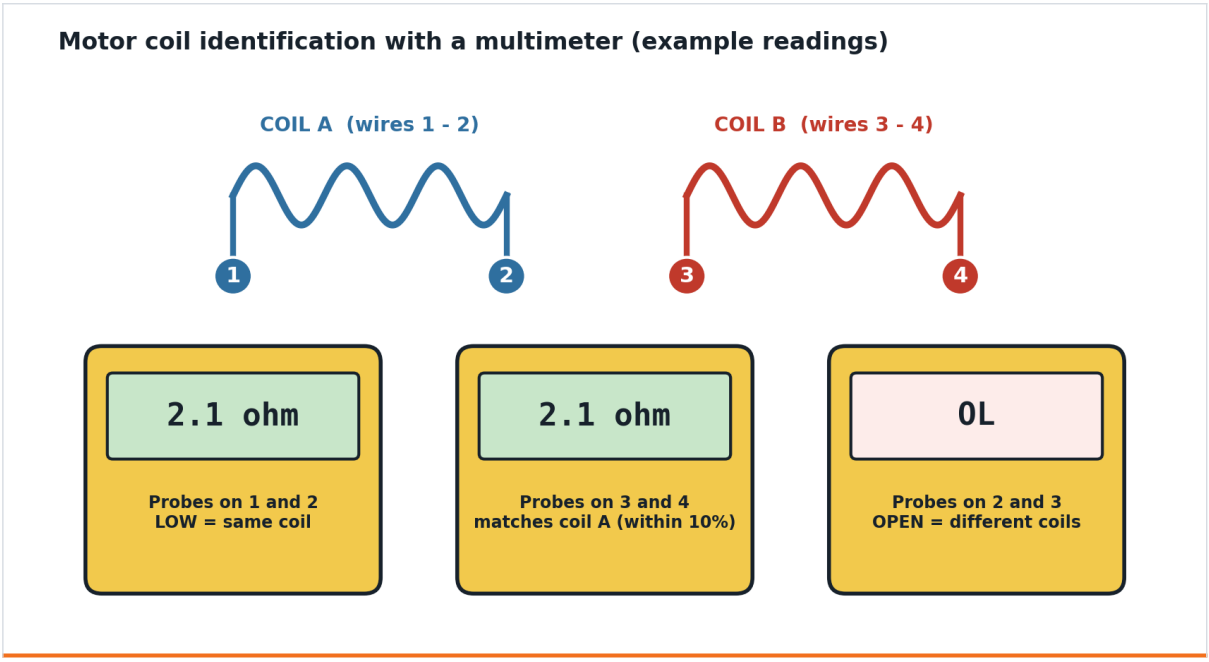


FIGURE 23 *Coil identification: low resistance inside a coil, open between coils*

Reading	Expected	Meaning if different
Coil A resistance	Matches coil B	Open circuit: broken winding or lead
Coil B resistance	Matches coil A	Much lower or higher: damaged winding
Coil A to coil B	Open circuit	Continuity: internal short between coils
Any coil to motor body	Open circuit	Continuity: insulation fault

11.3 Driver Current Setting

Set Vref for each driver using Section 6.4, with no motors connected, at 70 to 80% of the rated motor current. Record the value in the sheet in Appendix 15.11.

11.4 Powered Pre-Check Sequence

1	Power off, fit the drivers, connect both motors and the servo, connect USB, then power on.
2	In Inkscape open AxiDraw Control > Manual and walk the carriage a short distance in each direction.
3	Confirm that the head moves the direction commanded. If mirrored, reverse that motor plug or swap one coil pair.
4	Run a back-and-forth walk for 2 minutes. Listen for skipped steps, stalls or resonance.
5	Touch the motors after 10 minutes. They should be warm, not too hot to hold; reduce Vref if too hot.

11.5 Microstepping Reference

Resolution setting	Microstepping	Steps per revolution (1.8° motor)
High (default)	1/16	3200
Low	1/8	1600

The record sheet for the pre-check is in Appendix 15.11.

12. Calibration and Acceptance Testing

Test	Method	Pass criterion
Free movement	Move carriage by hand, power off	No binding or grinding over full travel
Squareness	Plot a 100 mm square	Diagonals equal within 1 mm
Pen lift	Toggle the pen 10 times	Full lift and lower every time
Scale	Plot a 200 mm line on each axis	Within 0.5% of nominal
Repeatability	Plot the same pattern twice	Lines overlap with no offset
Writing time	Plot one A4 page of cursive text	About 6 to 8 minutes [Spec]
Thermal	Run for 15 minutes	Motors warm, not too hot to touch

- Pen height: set pen-up and pen-down heights in AxiDraw Control > Setup before moving the servo horn.
- Belt tension: re-tension if lines show a double edge or shift at direction changes.
- Skew: loosen the end blocks and re-square the rod pairs if the square test fails.
- Half-size or double-size plots: check the Resolution setting (Section 6.3).

13. Operation and Maintenance

13.1 Installation Procedure

1	Unpack the machine and keep the box for transport.
2	Place it on a flat, stable, dry surface with free space around the head and about 56 × 40 cm of room.
3	Check that the carriage moves by hand over its full travel (power off) and the belt is intact.
4	Install Inkscape and the AxiDraw software (Section 10) from Chrome.
5	Connect the USB cable between the machine and the computer.
6	Connect the 12 V adapter to the machine last, then to the mains.
7	Confirm that the machine is detected using Extensions > AxiDraw Control.

WARNING Keep fingers, hair and loose cables clear of the carriage and belt. Use only the supplied 12 V 2 A adapter.

13.2 Initial Setup

1	Set Resolution to High: Extensions > AxiDraw Control > Options (Section 6.3).
2	Walk the carriage to the home corner and place A4 paper aligned with the corner.
3	Fit a pen (up to 15 mm diameter) in the holder and tighten the thumb screw.
4	In the Setup tab, set pen-up and pen-down heights so the pen clears the paper when up and just touches when down.
5	Set the page size in Inkscape to A4 (or within 225 × 320 mm).
6	Run a short test plot (for example a 50 mm square) and check size and pen lift.

13.3 Startup and Shutdown Procedure

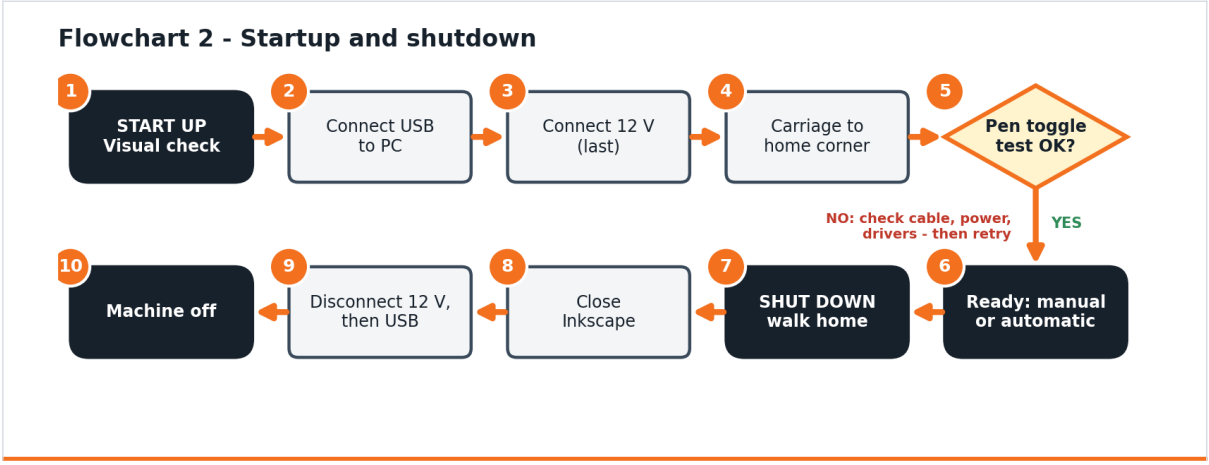


FIGURE 24 *Startup and shutdown flowchart*

Step	Startup
1	Visual check: paper, pen, cables, clear travel
2	Connect USB to the PC
3	Connect the 12 V adapter (last)
4	Carriage to the home corner by hand
5	Run a pen toggle test in AxiDraw Control
Shutdown	Raise the pen, walk home, close Inkscape, disconnect the 12 V adapter first and then USB

13.4 Manual Operation

Use the Manual tab of AxiDraw Control for movements that do not need a drawing file:

Function	Use
Walk X / Y	Move the head a set distance, for alignment and checks
Raise / lower pen	Test the servo and set pen height
Walk home	Return the head to the home corner
Disable motors	Free the motors so the carriage can be moved by hand
Pause button (board)	Pause a plot in progress

Always raise the pen before walking the head across the paper.

13.5 Automatic Operation

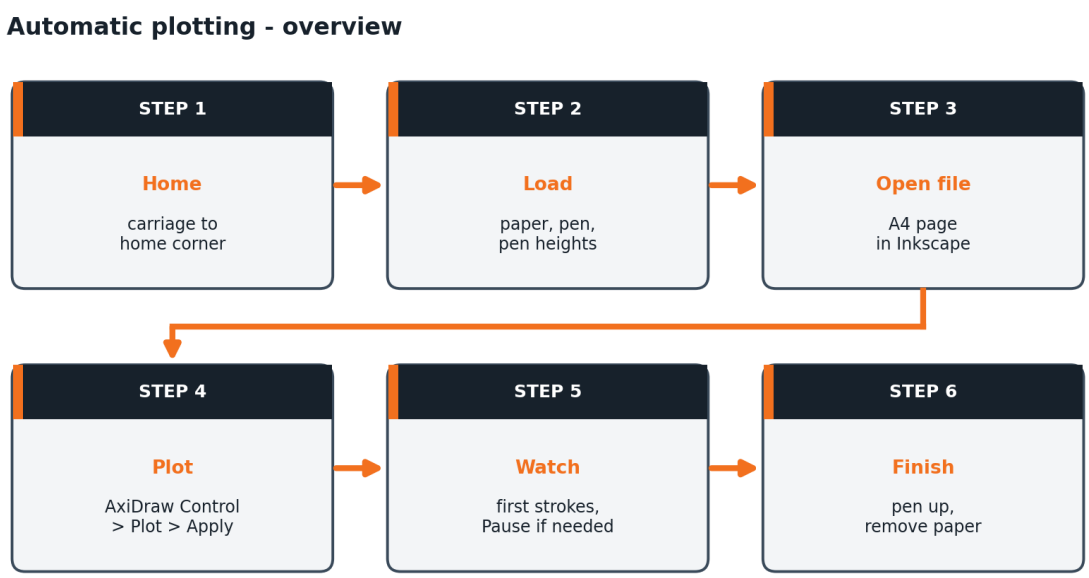


FIGURE 25 *Automatic plotting: step-by-step overview*

1	Open or create the drawing in Inkscape with an A4 page. Convert text to paths for text plots.
2	Open Extensions > AxiDraw Control.
3	In the Setup tab check pen-up and pen-down heights and speeds.
4	In the Plot tab press Apply. The machine writes and lifts the pen between strokes.
5	Watch the first strokes. Use the Pause button to stop if the pen skips or the paper moves.
6	When finished the head returns home. Raise the pen, remove the paper and power off.

NOTE Typical time for one A4 page: 6 to 8 minutes for cursive fonts, 8 to 20 minutes for non-cursive fonts.

13.6 Preventive Maintenance

Interval	Task
Before each use	Check the pen, paper, cables and that the carriage moves freely
Weekly	Wipe dust from rods, bearings and belt
Monthly	Check belt tension, screws and pulley grub screws; check driver temperature
Every 3 months	Re-check Vref; inspect printed parts for cracks or warping
Yearly	Inspect bearings, belt and pulleys for wear; replace as needed

13.7 Lubrication and Inspection

Item	Action
Guide rods	Wipe clean, then apply a thin film of light machine oil monthly; wipe off excess
Linear bearings	Slide along the oiled rods; do not over-oil
Belt	Do not lubricate; keep clean and dry
Motors, drivers	No lubrication; keep heatsinks clean and fit a fan above about 1 A
Printed parts	Inspect for cracks and warping

Inspection item	Check	Remarks
Belt condition and tension	☐	
Rods straight and clean	☐	
Bearings run smoothly	☐	
Screws and grub screws tight	☐	
Cables, plugs, JST-XH connectors	☐	
Servo horn and pen holder	☐	

13.8 Troubleshooting Guide

Symptom	Likely cause	Action
Inkscape cannot find the machine	USB cable is charge-only, no 12 V power, or driver missing	Use a data cable; connect the adapter; install the AxiDraw Windows installer
Motors do not move	No power, drivers fitted wrongly or loose plug	Check supply, driver EN marking and connectors
Drawing is half size	TMC driver with Resolution set to Low	Set Resolution to High (Section 6.3)
Drawing is mirrored	Motor direction reversed	Reverse that motor plug or swap one coil pair
Motor buzzes, no rotation	Coil pairs mixed or driver wrongly fitted	Re-check coil pairs and driver orientation
Motor runs hot	Vref too high	Lower Vref to 70 to 80% of rated current
Louder TMC2209 sound	Some brands run spreadCycle mode	Plots still draw correctly; try another driver brand for quieter running
Plot is distorted	Axes not square or loose belt	Re-square frame; re-tension belt
Pen does not lift	Servo horn slipped or heights wrong	Reset pen heights in the Setup tab; re-tighten the horn
Missed steps	Binding axis or drying rods	Lubricate rods; lower speed

13.9 Spare Parts List

No.	Part	Qty	Specification	Remarks
1	Cartizion Bot EBB Control Board	1	EBB firmware v3.0.3, 54 × 48 mm	Spare board
2	StepStick driver	2	A4988, TMC2208 or TMC2209	Fit with heatsink
3	Stepper motor	2	NEMA 17 bipolar	Match rated current
4	Pen-lift servo	1	MG90S-type 5 V micro servo	With horn
5	Timing belt	1	GT2 2 mm [Verify]	Cut to length
6	Guide rods	2 to 4	Smooth steel, 8 mm [Verify]	
7	Linear bearings	8	Match rod diameter	
8	ABS printed parts set	1	Carriage plates, mounts, pen head, covers	Contact Tharanee.dev
9	Power adapter	1	12 V 2 A, 5.5 × 2.1 mm	
10	USB cable	1	Type-A to Type-B	
11	Fasteners and clips	Set	Screws, nuts, black clips	

14. Project Summary and Future Development

14.1 Project Achievements

- A complete pen plotter built, assembled and documented, with a 225 × 320 mm drawing area for A4 paper
- An EBB-compatible controller that works with free, public software (Inkscape and the AxiDraw extension)
- Strong ABS structure made in a 70 °C heated chamber, plus injection-moulded parts
- Quiet running available with TMC2208 / TMC2209 drivers
- Plug-and-play delivery: install the software, connect USB and power, then draw
- A full technical document set: specification, assembly, pre-check, operation and maintenance

14.2 Technical Challenges and Solutions

Challenge	Solution
PLA parts are brittle and soften in heat	All structural parts printed in ABS in a 70 °C heated chamber
Aligning four bearings per axis so the carriage does not bind	Fit rods through the bearings before final screw tightening and tighten in a cross pattern
Driver noise and heat	TMC2208 / TMC2209 drivers, Vref at 70 to 80% of rated current, heatsinks and optional fan
Drawings half size or mirrored after driver changes	Keep Resolution on High for TMC drivers; reverse one motor plug if mirrored
Different driver types on one board	Standard StepStick sockets and a Vref table for each driver type
Many small loose parts during assembly	Sort fasteners and black clips on a sheet and assemble in the photographed order

14.3 Lessons Learned

- Check straightness of the rods and burrs on the printed parts before assembly; they decide how smoothly the carriage runs.
- Set the driver current before connecting motors; never plug or unplug a motor with power on.
- Record motor ratings and Vref values in the test report so later service starts from a known baseline.
- Keep all settings (Resolution, pen heights) written down after commissioning.
- Photograph every assembly stage; the photo index makes later repairs faster.

14.4 Future Improvements

- Optional X and Y limit switches for automatic homing
- Enclosure and cable chain for cleaner, safer cable routing
- Belt tensioners on the printed end blocks
- Paper clamps or a vacuum bed for sheet hold-down
- A 24 V variant using the board 12 to 24 V input for higher speed

- Extended drawing area for A3 sheets
- Pen-pressure adjustment and an automatic pen-change option

14.5 Possible Industrial Applications

- Training platform for motion control, stepper drives and belt kinematics
- Handwritten-style letters, labels and certificates
- Marking and labelling of flat parts with a pen or marker
- Testing touchscreens and touch devices with a stylus
- Plotting engineering drawings and layouts on A4 sheets
- Pen-based artwork and low-volume print-on-demand drawings

14.6 Conclusion

The Cartizion Bot V1.1 combines a rigid ABS-printed CoreXY-style frame, an EBB controller with removable stepper drivers, and free Inkscape software into a ready-to-use pen plotter. This document provides the technical basis to build, commission, operate and maintain the machine, and the appendices give the data needed for procurement, service and quality records.

15. Appendices

15.1 Complete Mechanical BOM

No.	Item	Qty	Material	Notes
1	Carriage plate, upper	1	ABS	Printed [Photo]
2	Carriage plate, lower (slotted)	1	ABS	Printed [Photo]
3	Motor bracket	2	ABS	One per motor
4	Rod end blocks / holders	4+	ABS	Hold rod ends rigid
5	Pen head housing and cradle	1	ABS	Holds servo and pen
6	Pen clamp and thumb screw	1	ABS / steel	Up to 15 mm pens
7	Covers (slotted)	2+	ABS	Electronics and belt run
8	Guide rods	4 to 6	Ground steel	8 mm typical [Verify]
9	Linear bearings	8	Bronze or steel	Match rod diameter
10	Timing belt	1	Rubber / glass fibre	GT2 [Verify]
11	Motor pulleys	2	Aluminium	20 teeth typical [Verify]
12	Idler pulleys	2+	Plastic or metal	At far end of belt run
13	Black clips and holders	~12	Plastic	Bearing and belt holders
14	Screws, nuts, washers	Set	Steel	M3 typical [Verify]

15.2 Electrical BOM

No.	Item	Qty	Specification	Notes
1	Cartizion Bot EBB Control Board	1	PIC18F46J50, fw v3.0.3, 54 × 48 mm	12 to 24 V input
2	StepStick driver	2	A4988, TMC2208 or TMC2209	Heatsinks fitted
3	Stepper motor	2	NEMA 17 bipolar	JST-XH 4-pin
4	Pen-lift servo	1	5 V micro servo	3-pin S / + / –
5	Power adapter	1	12 V 2 A, 5.5 × 2.1 mm	24 V 3 A for 24 V builds
6	USB cable	1	Type-A to Type-B	
7	Heatsinks / fan	2 / 1	For StepStick drivers	Fan above about 1 A

15.3 2D Manufacturing Drawings

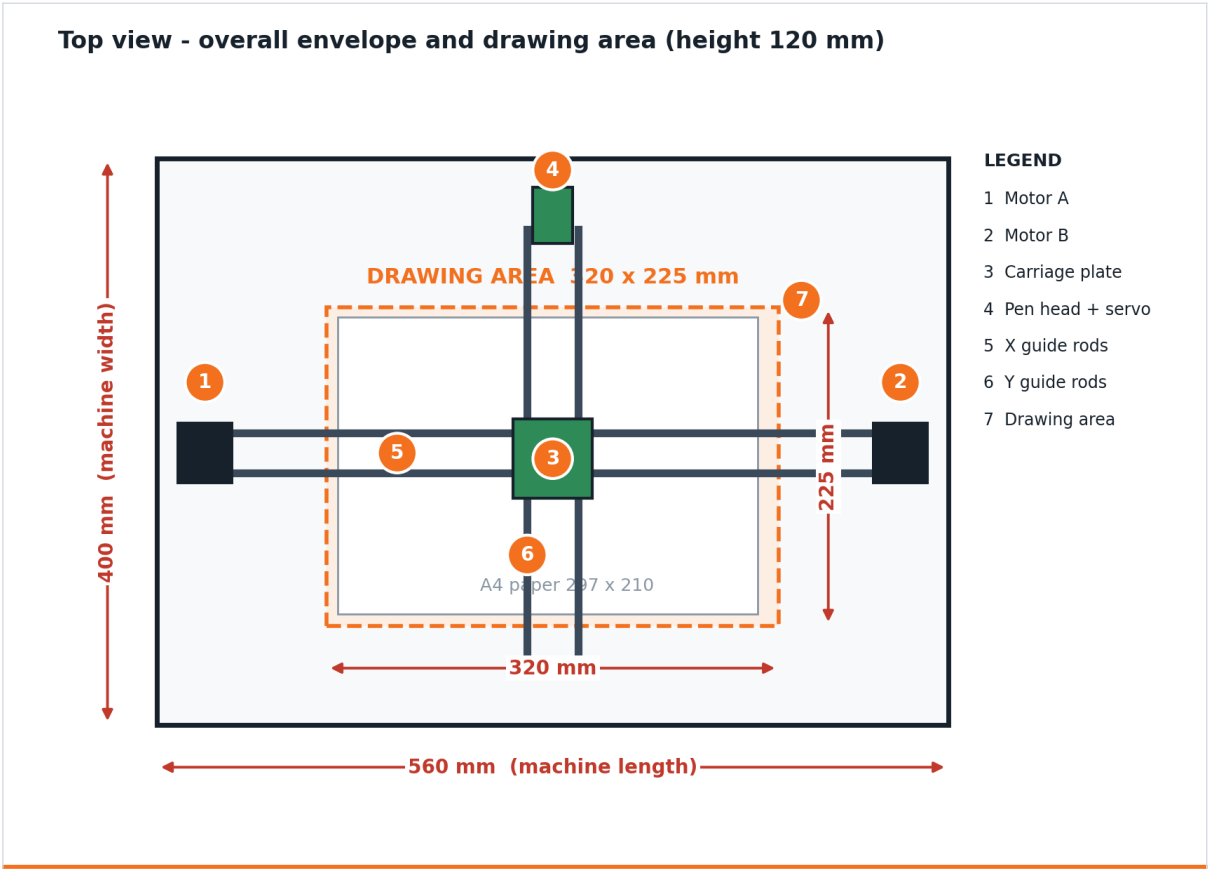


FIGURE 26 *Top view: machine envelope and drawing area*

Dimension	Value
Machine length × width × height	560 × 400 × 120 mm [Spec]
Drawing area	320 mm (long axis) × 225 mm
A4 paper	297 × 210 mm, placed inside the drawing area
Manufacturing files	Printed-part STL / STEP and dimensioned sheets: [attach from the CAD project]

ABS printing parameter	Value
Material	ABS
Chamber temperature	70 °C heated chamber [Spec]
Printer type	Enclosed, actively heated printer (water-cooled hot end) [Spec]
Nozzle, layer height, infill	[Set per part; 40% or higher infill on load-bearing parts, Verify]

15.4 3D CAD Assembly Views

Native 3D CAD files are not included in this revision. The assembly views below are taken from the build photographs and the contact sheets index all 56 photos in time order.

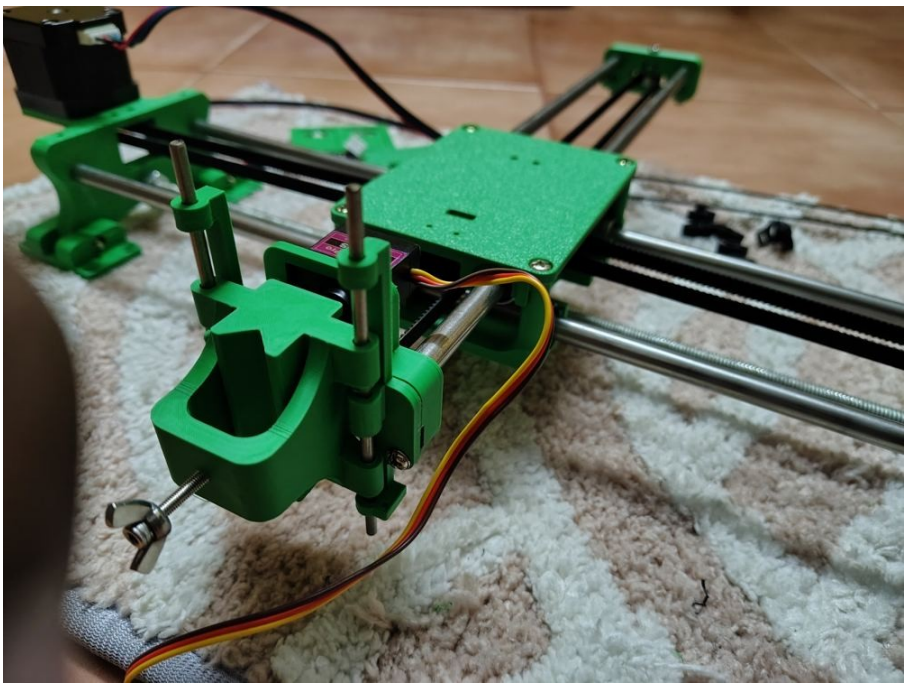


FIGURE 27 *Assembly view: pen head and carriage*

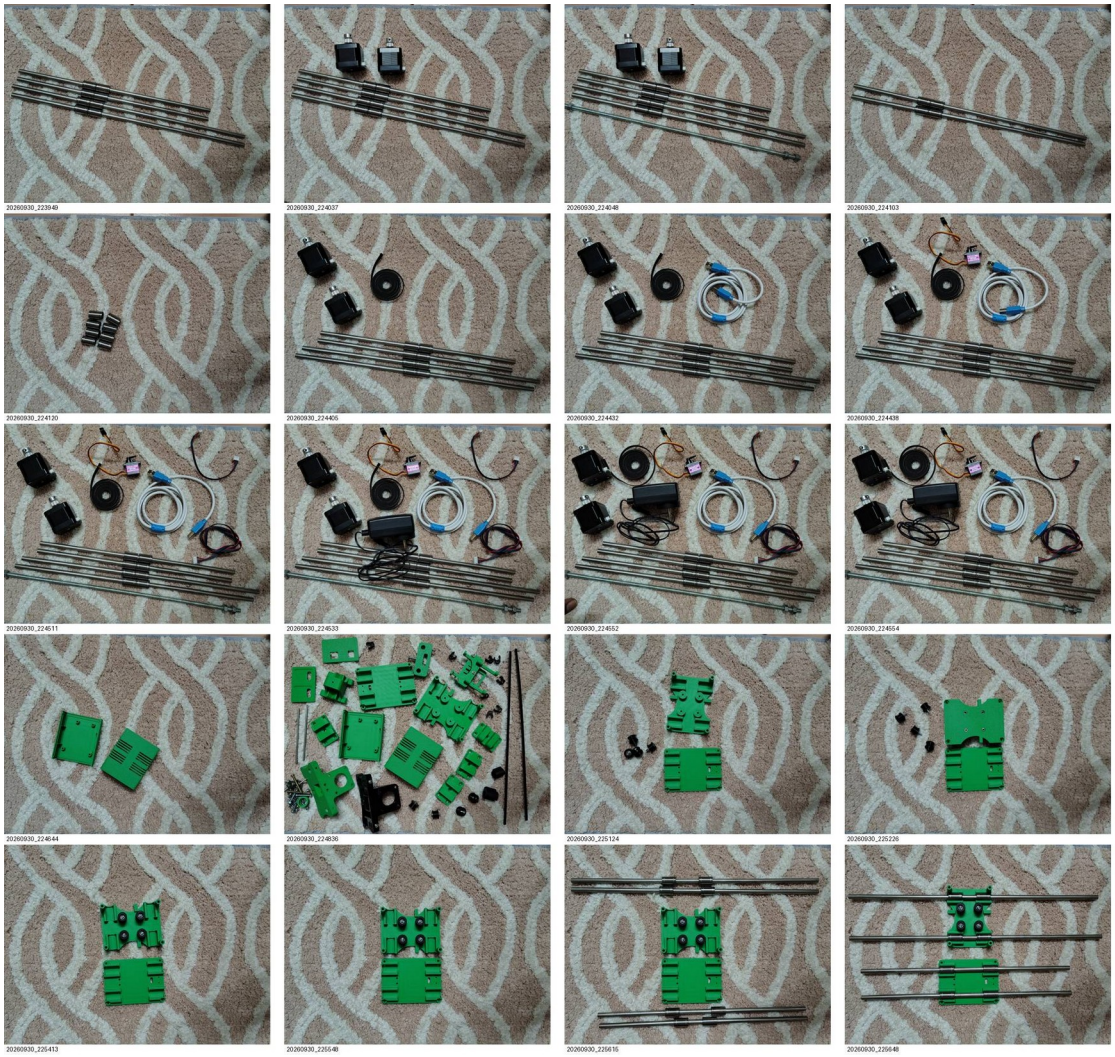


FIGURE 28 *Photo index 1: parts, kit and carriage plate (22:39 to 22:56)*

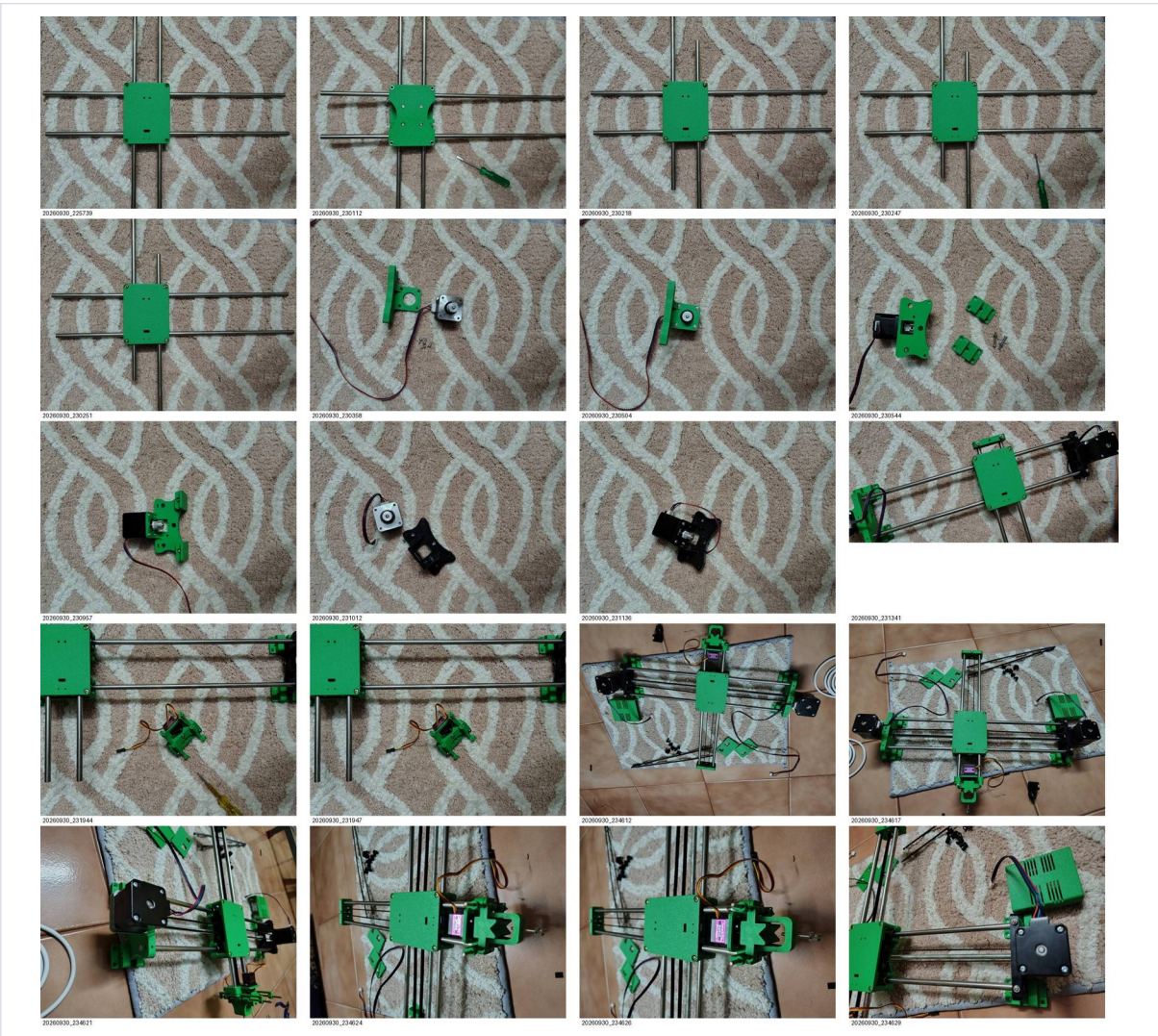


FIGURE 29 *Photo index 2: carriage, motor brackets and pen head (22:57 to 23:46)*



FIGURE 30 *Photo index 3: belt, frame and final layout (23:46 to 23:49)*

15.5 Electrical Schematics

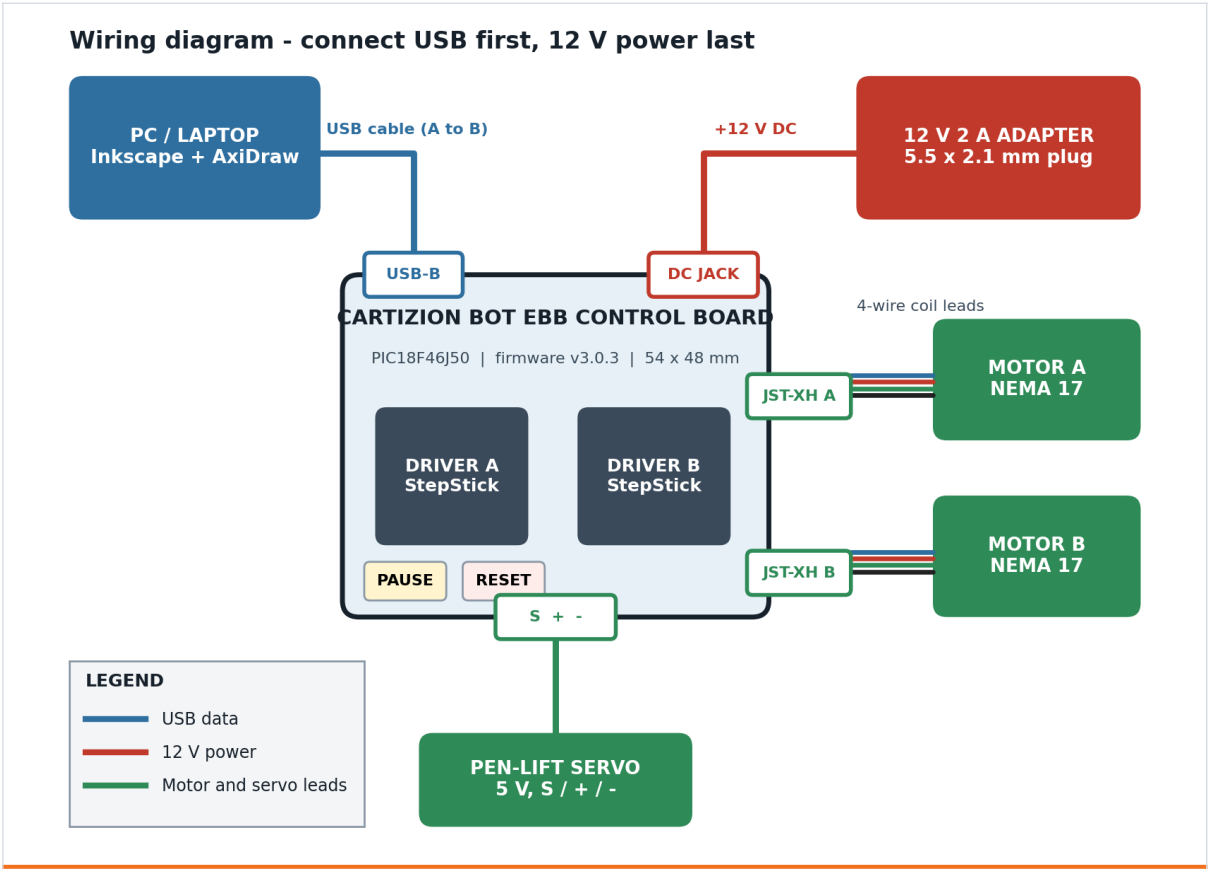


FIGURE 31 *Wiring diagram: PC, adapter, EBB board, motors and servo*

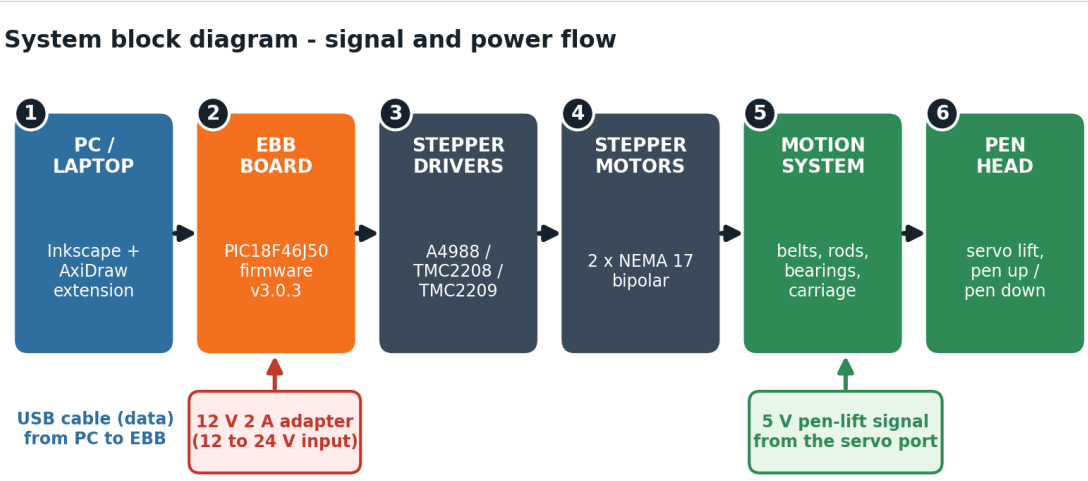


FIGURE 32 *System block diagram: signal and power flow*

Signal	From	To
USB data and 5 V logic	PC USB-A	Board USB-B
12 V DC	Adapter barrel plug	Board DC jack
Motor coils (A+, A−, B+, B−)	StepStick driver A / B	Motor JST-XH plug
Servo signal, +5 V, GND	Board S / + / − header	Pen-lift servo

15.6 Controller I/O Mapping

Function	Board connection	Notes
Power input	DC barrel jack 5.5 × 2.1 mm	12 to 24 V DC
Computer link	USB Type-B	EBB firmware v3.0.3
Motor A	JST-XH 4-pin	Driver socket A
Motor B	JST-XH 4-pin	Driver socket B
Pen lift	3-pin S / + / – header	5 V servo
Pause	Push button	Pauses a running plot
Reset	Push button	Resets the board

Optional Arduino UNO + CNC shield bench map

Used only by the reference sketches in Section 15.10.

Function	Pin	Notes
Motor A step / dir	D2 / D5	CNC shield V3
Motor B step / dir	D3 / D6	CNC shield V3
Driver enable	D8	LOW = enabled
Pen servo	D11	PWM pin

15.7 Motor and Drive Datasheets

Typical datasheet values. Replace [Verify] entries with the values printed on the fitted parts and attach the manufacturer datasheets.

Parameter	Stepper motor NEMA 17 [Verify]	Fitted motor (record)
Model / marking		_____
Rated current (A)	1.0 to 1.5	_____
Coil resistance (Ω)	1.5 to 3.5	_____
Inductance (mH)	2 to 6	_____
Holding torque (N·m)	0.35 to 0.55	_____
Step angle	1.8°	_____
Shaft	5 mm D-cut	_____

Driver	Supply (V)	Current	Microsteps	Remarks [Verify datasheet]
A4988	8 to 35	1 A without cooling, up to 2 A with cooling	Up to 1/16	Sense R100 or R068; Vref = I × 0.8 or × 0.54

Driver	Supply (V)	Current	Microsteps	Remarks [Verify datasheet]
TMC2208	4.75 to 36	1.2 A RMS, 2 A peak	Up to 1/16, interpolated to 1/256	Near-silent; Vref = I × 1.0 (R110)
TMC2209	4.75 to 29	2 A RMS, 2.8 A peak	Up to 1/16, interpolated to 1/256	Near-silent; Vref = I × 1.0 (R110)

15.8 Sensor Datasheets

The standard machine has no sensors. Homing is done by placing the carriage at the home corner by hand, and the board has Pause and Reset buttons. Optional limit switches (Section 14.4) would use the following reference specification.

Item	Typical specification [Verify]
Mechanical limit switch	Normally open or closed micro-switch with lever, 5 V logic compatible
Mounting	Printed bracket on the rod end block
Wiring	Two wires to the controller input and ground

15.9 Calculation Sheets

Steps per millimetre (belt drive)

NOTE steps per mm = (200 × microstepping) ÷ (belt pitch × pulley teeth)

Resolution	Steps per revolution	Steps per mm	Resolution (mm)
Low (1/8)	1600	40	0.025
High (1/16)	3200	80	0.0125

Example: 200 steps × 16 = 3200 steps per revolution; one revolution moves 20 × 2 mm = 40 mm; so 3200 ÷ 40 = 80 steps per mm (about 2032 steps per inch) [Verify belt and pulley].

Speed and step rate

Step rate = speed × steps per mm. At the 330 mm/s maximum, with 80 steps per mm: 330 × 80 = 26,400 steps per second per motor for an axis-aligned move. Motor speed = 330 ÷ 40 = 8.25 rev/s = 495 rpm.

Belt force

Pulley pitch radius = (20 × 2 mm) ÷ (2π) = 6.37 mm. With a typical holding torque of 0.4 N·m, belt force = 0.4 ÷ 0.00637 = about 63 N. At 75% current about 47 N. The carriage and pen head need only a few newtons, so the motors have a large margin.

Vref examples

Case	Calculation
A4988 R100, motor 1.2 A at 75% = 0.9 A	Vref = 0.9 × 0.8 = 0.72 V

Case	Calculation
A4988 R068, motor 1.2 A at 75% = 0.9 A	Vref = 0.9 × 0.54 = 0.49 V
TMC2209 R110, motor 1.5 A at 75% = 1.1 A	Vref = 1.1 × 1.0 = 1.10 V

CoreXY kinematics

For a CoreXY belt drive the two motors move A = X + Y and B = X − Y (in belt millimetres), and the head position is X = (A + B) ÷ 2, Y = (A − B) ÷ 2. If only one motor turns the head moves diagonally. The EBB and AxiDraw software apply this conversion; it is needed only when writing your own controller.

Electrical power

Supply power = 12 V × 2 A = 24 W maximum. Writing time estimate: 6 to 8 minutes per A4 page for cursive text [Spec].

15.10 Software Flowcharts and Program References

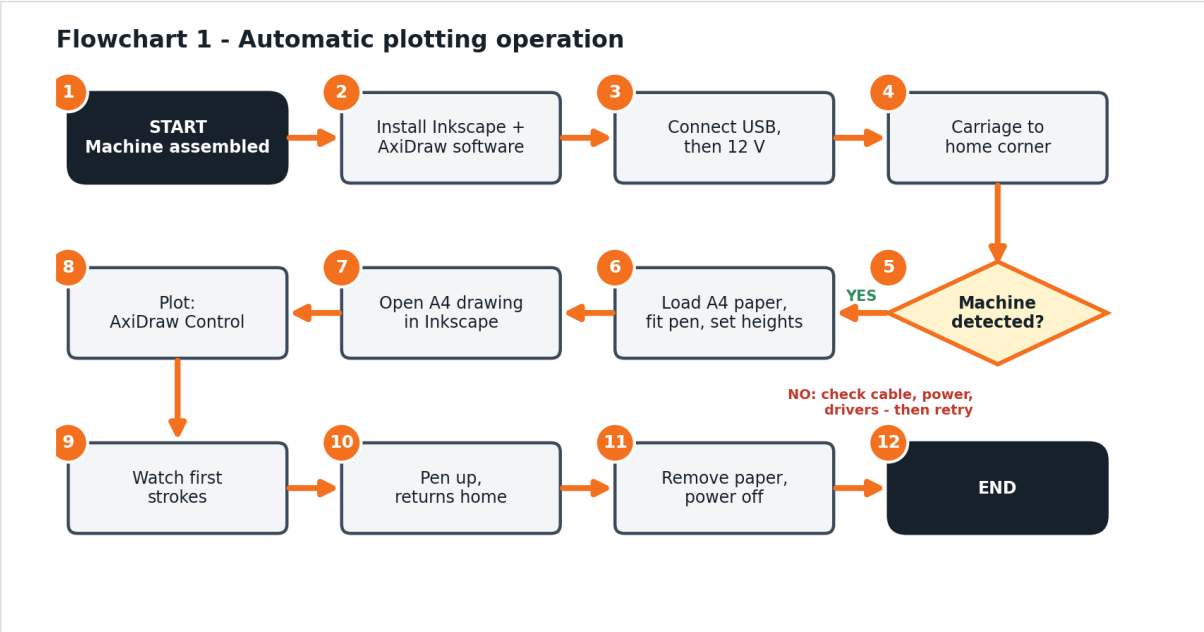


FIGURE 33 *Flowchart 1: automatic plotting operation*

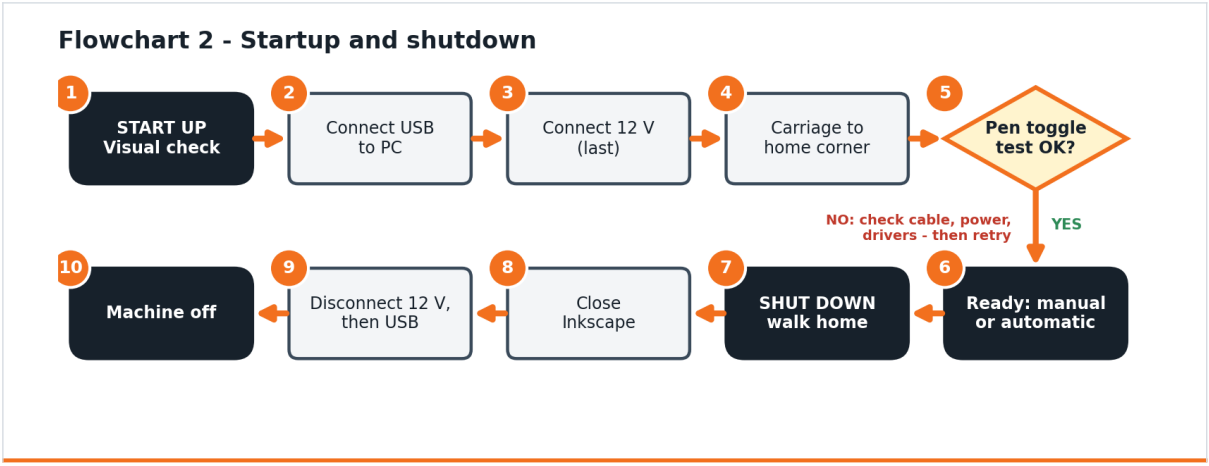


FIGURE 34 *Flowchart 2: startup and shutdown*

The Cartizion Bot V1.1 needs no code to be uploaded: the controller is ready with EBB firmware v3.0.3. The reference sketches below are optional, for bench tests of motors and the servo with an Arduino UNO + CNC shield and A4988 drivers (never connected at the same time as the EBB board). Sketch 3 includes a CoreXY option.

</> SKETCH 1 - stepper_test.ino (motor A / motor B test)

```
const int A_STEP = 2, A_DIR = 5;
const int B_STEP = 3, B_DIR = 6;
const int EN_PIN = 8;
const int MICROSTEP = 8;               // match jumpers (tune)
const long STEPS_PER_REV = 200L * MICROSTEP;
const unsigned int STEP_US = 400;      // half-period in microseconds (tune)

void setup() {
  Serial.begin(115200);
  pinMode(A_STEP, OUTPUT); pinMode(A_DIR, OUTPUT);
  pinMode(B_STEP, OUTPUT); pinMode(B_DIR, OUTPUT);
  pinMode(EN_PIN, OUTPUT);
  digitalWrite(EN_PIN, LOW);           // enable drivers
  Serial.println(F("a/A = motor A fwd/back, b/B = motor B fwd/back, t = 2 min test"));
}

void moveSteps(int stepPin, int dirPin, bool fwd, long n) {
  digitalWrite(dirPin, fwd ? HIGH : LOW);
  for (long i = 0; i < n; i++) {
    digitalWrite(stepPin, HIGH);
    delayMicroseconds(STEP_US);
    digitalWrite(stepPin, LOW);
    delayMicroseconds(STEP_US);
  }
}

void loop() {
  if (!Serial.available()) return;
  char c = Serial.read();
  switch (c) {
    case 'a': moveSteps(A_STEP, A_DIR, true,  STEPS_PER_REV); break;
    case 'A': moveSteps(A_STEP, A_DIR, false, STEPS_PER_REV); break;
    case 'b': moveSteps(B_STEP, B_DIR, true,  STEPS_PER_REV); break;
    case 'B': moveSteps(B_STEP, B_DIR, false, STEPS_PER_REV); break;
    case 't': {
      unsigned long t0 = millis();
      while (millis() - t0 < 120000UL) {
        moveSteps(A_STEP, A_DIR, true,  STEPS_PER_REV);
        moveSteps(A_STEP, A_DIR, false, STEPS_PER_REV);
        moveSteps(B_STEP, B_DIR, true,  STEPS_PER_REV);
        moveSteps(B_STEP, B_DIR, false, STEPS_PER_REV);
      }
      Serial.println(F("Test complete"));
      break;
    }
    default: return;
  }
  Serial.println(F("done"));
}
```

</> SKETCH 2 - pen_servo_test.ino

```
#include <Servo.h>
const int PEN_PIN = 11;
Servo pen;

void setup() {
  Serial.begin(115200);
  pen.attach(PEN_PIN);
  pen.write(90);
  Serial.println(F("Send angle 0-180 to move the pen servo"));
}

void loop() {
  if (Serial.available()) {
    int angle = Serial.parseInt();
    if (angle >= 0 && angle <= 180) {
      pen.write(angle);
      Serial.print(F("angle = "));
      Serial.println(angle);
    }
  }
}
```

</> SKETCH 3 - xy_controller.ino (G-code subset, CoreXY or Cartesian)

```
// Supported: G0 G1 G90 G91 G92 G28 M3 M5 M114 (mm, feed in mm/min)
#include <Servo.h>
#define A_STEP 2
#define A_DIR  5
#define B_STEP 3
#define B_DIR  6
#define EN_PIN 8
#define PEN_PIN 11

const bool  COREXY = true;             // false = Cartesian
const float SPM = 80.0;                // steps per mm of belt travel (tune)
const float X_MAX = 320.0, Y_MAX = 225.0;
const bool  INVERT_A = false, INVERT_B = false;
const int   PEN_UP_ANGLE = 90, PEN_DOWN_ANGLE = 55;   // from Sketch 2
const unsigned long MIN_STEP_US = 200;

Servo pen;
long posA = 0, posB = 0;               // motor positions in steps
bool absMode = true;
float feedRate = 1200.0;
String line = "";

void toMotors(float x, float y, long &a, long &b) {
  if (COREXY) { a = lround((x + y) * SPM); b = lround((x - y) * SPM); }
  else        { a = lround(x * SPM);       b = lround(y * SPM); }
}
void toXY(long a, long b, float &x, float &y) {
  if (COREXY) { x = (a + b) / (2.0 * SPM); y = (a - b) / (2.0 * SPM); }
  else        { x = a / SPM;               y = b / SPM; }
}
void waitUs(unsigned long us) {
  if (us > 16000UL) { delay(us / 1000UL); delayMicroseconds(us % 1000UL); }
  else delayMicroseconds(us);
```

```
}
void penUp()   { pen.write(PEN_UP_ANGLE);   delay(150); }
void penDown() { pen.write(PEN_DOWN_ANGLE); delay(150); }

void moveTo(float x, float y, float f) {
  x = constrain(x, 0.0, X_MAX);
  y = constrain(y, 0.0, Y_MAX);
  long ta, tb;
  toMotors(x, y, ta, tb);
  long da = ta - posA, db = tb - posB;
  long aa = labs(da), ab = labs(db);
  long n = max(aa, ab);
  if (n == 0) return;
  digitalWrite(A_DIR, ((da >= 0) != INVERT_A) ? HIGH : LOW);
  digitalWrite(B_DIR, ((db >= 0) != INVERT_B) ? HIGH : LOW);

  float cx, cy;
  toXY(posA, posB, cx, cy);
  float distMM = sqrt(sq(x - cx) + sq(y - cy));
  float seconds = distMM / (f / 60.0);
  unsigned long periodUs = (unsigned long)((seconds * 1000000.0) / n);
  if (periodUs < MIN_STEP_US) periodUs = MIN_STEP_US;

  long ea = 0, eb = 0;
  for (long i = 0; i < n; i++) {
    ea += aa; eb += ab;
    bool sa = false, sb = false;
    if (ea >= n) { ea -= n; sa = true; }
    if (eb >= n) { eb -= n; sb = true; }
    if (sa) digitalWrite(A_STEP, HIGH);
    if (sb) digitalWrite(B_STEP, HIGH);
    delayMicroseconds(5);
    digitalWrite(A_STEP, LOW);
    digitalWrite(B_STEP, LOW);
    waitUs(periodUs - 5);
  }
  posA = ta; posB = tb;
}

float getVal(const String &s, char key, float def, bool &found) {
  int i = s.indexOf(key);
  if (i < 0) { found = false; return def; }
  found = true;
  return s.substring(i + 1).toFloat();
}

void process(String s) {
  s.trim(); s.toUpperCase();
  int sc = s.indexOf(';');
  if (sc >= 0) s = s.substring(0, sc);
  s.trim();
  if (s.length() == 0) { Serial.println(F("ok")); return; }
  char type = s.charAt(0);
  int code = s.substring(1).toInt();
  float curX, curY;
  toXY(posA, posB, curX, curY);
  bool fx, fy, ff;
  float vx = getVal(s, 'X', curX, fx);
```

```
  float vy = getVal(s, 'Y', curY, fy);
  float vf = getVal(s, 'F', feedRate, ff);
  if (ff) feedRate = vf;

  if (type == 'G') {
    if (code == 0 || code == 1) {
      float tx = absMode ? vx : curX + (fx ? vx : 0);
      float ty = absMode ? vy : curY + (fy ? vy : 0);
      moveTo(tx, ty, code == 0 ? 3000.0 : feedRate);
    }
    else if (code == 90) absMode = true;
    else if (code == 91) absMode = false;
    else if (code == 92) { if (fx || fy) toMotors(vx, vy, posA, posB); }
    else if (code == 28) { penUp(); moveTo(0, 0, 3000.0); }
    else { Serial.println(F("error: unknown G")); return; }
  }
  else if (type == 'M') {
    if (code == 3) penDown();
    else if (code == 5) penUp();
    else if (code == 114) {
      Serial.print(F("X:")); Serial.print(curX, 2);
      Serial.print(F(" Y:")); Serial.println(curY, 2);
    }
    else { Serial.println(F("error: unknown M")); return; }
  }
  else { Serial.println(F("error: unknown command")); return; }
  Serial.println(F("ok"));
}

void setup() {
  Serial.begin(115200);
  pinMode(A_STEP, OUTPUT); pinMode(A_DIR, OUTPUT);
  pinMode(B_STEP, OUTPUT); pinMode(B_DIR, OUTPUT);
  pinMode(EN_PIN, OUTPUT);
  digitalWrite(EN_PIN, LOW);
  pen.attach(PEN_PIN);
  penUp();
  Serial.println(F("PLOTTER READY"));
}

void loop() {
  while (Serial.available()) {
    char c = Serial.read();
    if (c == '\n' || c == '\r') {
      if (line.length()) { process(line); line = ""; }
    } else line += c;
  }
}
```

NOTE These sketches are untested bench references. Verify steps per mm, pen angles, motor direction and CoreXY sign before connecting a pen. A UNO reaches roughly 5,000 to 10,000 steps per second, below the EBB board capability.

Other program references

Reference	Use
Inkscape + AxiDraw extension	Plot, Setup, Options, Manual tabs (Sections 10 and 13)
axidraw_conf.py	Set resolution = 1 for High resolution in command-line / Python use
EBB firmware v3.0.3	Update over USB without a programmer
axidraw.com/sw	Official AxiDraw software installers and instructions

15.11 Test Reports

Stepper motor pre-check record

Check	Motor A	Motor B	Pass criterion
Visual and shaft spin OK	☐	☐	No damage, smooth rotation
Coil 1 resistance (Ω)			Matches coil 2 within 10%
Coil 2 resistance (Ω)			Matches coil 1 within 10%
Coil to body open	☐	☐	No continuity
Driver type and Vref (V)			Section 6.4 formula
Direction correct	☐	☐	Not mirrored
2 min back-and-forth test	☐	☐	No skipped steps
Temperature after 10 min			Warm, below about 60 °C
Inspector / date			

Acceptance test report

Test	Criterion	Measured	Pass / Fail
Free movement	No binding		
100 mm square diagonals	Within 1 mm		
200 mm line, X axis	Within 0.5%		
200 mm line, Y axis	Within 0.5%		
Pen lift ×10	All OK		
A4 cursive page time	6 to 8 min		
Thermal 15 min	Warm only		

15.12 Revision History

Rev.	Date	Description	Author
A	29 Oct 2026	Initial draft of the Cartizion Bot V1.1 product technical document	Tharanee.dev

Approval	Name	Signature	Date
Prepared			
Reviewed			
Approved			